

Master Thesis

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# **A Pyramidal Neuron's Account of Predictive Learning: Novelty and Omission Signals without Prediction Error**

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by

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# A Pyramidal Neuron’s Account of Predictive Learning: Novelty and Omission Signals without Prediction Error

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## ABSTRACT

Animals immersed in dynamic environments require neural mechanisms for surprise and anticipation - grounded in, and acquired through experience. Popular predictive coding theories of neural function argue that such abilities arise from a generative process where brains continuously compute prediction errors by comparing their internal predictions with external inputs, and locally refine their internal connectivity to minimize these errors. However, key assumptions of biologically-grounded implementations of predictive coding lack anatomical support. Most evidently, top-down recruitment of interneuron types required for calculating prediction errors is sparse in superficial layers of the cortex, which are assumed to be responsible precisely for this computation. This work asks how predictive learning could occur in L2/3 circuits consistent with our knowledge of cortical anatomy - forgoing normative assumptions about feedback-driven inhibition, and distinct populations of prediction error cells. Combining data analysis of experimental evidence from a visual occlusion paradigm, and data-driven computational modeling, we propose a context-contrasting circuit model of predictive learning in L2/3 pyramidal cells. Plasticity in our circuit relies on opposing learning rules in apical and somatic compartments of pyramidal cells, resulting in experience-dependent strengthening of context-driven apical activity and weakening of sensory-driven somatic activity, consistent with prior work. The context-contrasting model allows us to reproduce the dominant modes of plasticity that emerge under visual occlusion, including response transitions structurally incompatible with prediction error circuits. Through in-silico silencing experiments, we isolate parts of our model responsible for the diverse types of plasticity observed in experimental data, leading to concrete hypotheses for future experiments. Together, our work provides a minimal account of local unsupervised learning of novelty and omission signals in L2/3 pyramidal cells, and contributes to development of anatomically-consistent theories of predictive learning in biological neural circuits.

## 1 INTRODUCTION

An organism that can only react to the past is always late, left lagging behind its ever-changing environment. Tracking, evading and intercepting other dynamic organisms or objects are key survival abilities for animals, impossible without internal anticipatory mechanisms for predicting future states of the environment based on prior experience [1, 2]. Neuroscience clearly demonstrates the existence of such predictive computations in the brain, required for rapid perception, action and decision-making [1]. The evidence ranges from contextual visual information influencing activity in early sensory regions [3], to neural responses to identical auditory

inputs being significantly reduced in a predictable compared to unexpected contexts [4], and existence of semantic and temporal hierarchy for speech prediction [5]. Still, recent progress in machine learning represents perhaps the most convincing demonstration of the computational value of prediction: modern artificial neural networks acquire general models of the world useful across many tasks without externally-provided labels or reward, learning purely by optimizing prediction of full sensory inputs from partial ones [6], learning to predict the same latent representations across different views of the same scene [7, 8], predicting future observations or latent representations from current ones [2, 9], or by learning to predict consequences of a planned and executed sequence of actions on future latent representations, enabling long-horizon planning [10]. Altogether, the computational benefit and biological prevalence of predictive computation and learning in the brain are solidly established [1, 11–13]. *How* the brain implements these general computational principles is much less clear, as the same functional outcomes can often be explained by a variety of neural architectures and learning algorithms [13].

The most mature, and widely-accepted account of predictive learning and computation in the brain is hierarchical predictive coding (hPC). Rao and Ballard [14] framed the visual cortex as a hierarchical generative model, where each level tries to explain its inputs in terms of hypothetical causes. hPC proposed that at each level of the hierarchy, feedback connections carry predictions of lower-level activity while feedforward connections carry the residual prediction errors between top-down predictions and bottom-up sensory input. Hierarchical neural networks implementing this scheme, trained on natural images, developed simple-cell-like classical receptive fields, and the units carrying prediction errors exhibited end-stopping (surround suppression) and other extra-classical receptive field effects as emergent consequences of subtractive top-down feedback. One of the core claims of hPC is that each layer of the cortical hierarchy contains two distinct types of cells - backward-projecting "representation" neurons carrying each layer's prediction of the latent causes of the visual scene, and forward-projecting "error" neurons conveying the residual between the bottom-up evidence and top-down prediction. Later work [15, 16] mapped hPC theory onto canonical cortical microcircuits with superficial L2/3 pyramidal cells (PyCs) identified as prediction-error neurons, and deep L5 PyCs as representation neurons. The notion of prediction-error neurons was further refined, to highlight that in biological implementation of the hPC scheme, the firing of a single prediction-error neuron cannot signal both a lack of prediction relative to sensory evidence (positive prediction error, PPE) and an excess of prediction relative to sensory evidence (negative prediction error, NPE), as the baseline firing rates in L2/3 PyCs are low [17]. It was therefore

proposed that the superficial L2/3 contains two distinct populations of positive and negative prediction-error neurons - both involved in steering the activity of the deep L5 representation neurons carrying each hierarchical level's estimates of the latent causes.

Since its introduction almost three decades ago, experimental neuroscientists have looked for empirical support for hPC in the brain. To date, some of the most convincing experimental evidence for hPC has come from studies of visuomotor coupling and mismatch in primary visual cortex (V1) [18–20], which compare responses of V1 neurons to artificially-controlled visual flow in a virtual tunnel versus locomotion-predicted flow based on the mouse's current running speed. The early experiments found that L2/3 neurons in V1 do not respond to visual flow, but rather to mismatches between visual and locomotion-predicted optic flow [18]; and subsequent whole-cell recordings showed that bottom-up visual flow and top-down locomotion-predicted flow exert influences of opposite sign on membrane potential of two distinct classes of L2/3 neurons, consistent with the interpretation that L2/3 consists of two populations of PPE and NPE neurons, while no signature of prediction error was found in L5 "representation" neurons [19]. The strongest case for L2/3 encoding of prediction errors by distinct PPE and NPE populations came from activity-dependent tagging and subsequent RNA sequencing of L2/3 cells during the visuomotor mismatch task, revealing the Rrad population of L2/3 PyC's to be enriched for PPE responses and the Adamts2 population of L2/3 PyC's to be enriched for NPE responses [20]. More recently, in auditory cortex, omission responses were found to scale with the amount of accumulated evidence for top-down predictions - supporting the role of L2/3 PyCs in signaling of graded error signals useful for predictive learning and inference beyond vision [21].

Despite the maturity of predictive-coding as a general theory of neural function, the empirical support for several crucial postulates of hPC has increasingly come under scrutiny. A review of the connectivity and interaction between sensory cortices suggests that feedback predominantly amplifies predicted representations, and that weakening of predictable stimuli can largely be attributed to lateral inhibition and adaptation of feedforward responses [22]. Indeed a growing amount of evidence [23, 24] challenges the ubiquity of top-down inhibition, required for PPE signaling in L2/3 neurons according to hPC [15, 17]. A detailed optogenetics-assisted circuit mapping of feedback inputs into V1 found exactly the top-down activation of SST and PV interneurons - required for PPE computation - to be missing from L2/3 [23], the layer where prediction errors are assumed to occur. Instead, top-down PyC amplification and disinhibition via VIP interneurons emerged as the dominant feedback motifs in superficial L2/3, while top-down inhibition via PV and SST interneurons was only observed in deep L5 and L6 layers [23]. Another recent series of optogenetics experiments in mice and monkeys during the local-global oddball paradigm found no evidence for top-down suppression of predictable responses or release from suppression for unpredictable responses through L2/3 interneurons [24]. Instead, suppression of locally-predictable responses was consistent with feedforward adaptation, and locally-unpredictable oddball responses were attributable to release from feedforward adaptation [24].

Another conceptual issue that has emerged in both computational neuroscience and machine learning concerns the hPC assumption that the brain is a generative model, with early sensory areas receiving generative top-down predictions about low-level details of the outside world. A growing body of theoretical [2, 11, 25, 26] and empirical [24, 27–29] evidence now suggests that the brain might predict at a higher level of abstraction than the generative hPC perspective suggests. Firing rates in awake macaque V1 are modulated mainly by contextual predictability of higher-order image features, as opposed to low-level features [27]. The same finding generalizes across the mouse visual system, with the effects strongest in superficial layers, traditionally thought to convey prediction errors [28]. The local-global oddball studies in mice and monkeys also support this conclusion - global oddball responses are restricted to neurons in higher-order cognitive areas outside the visual cortex, and these are the only regions showing clear signatures of single-neuron prediction-error computations [24, 29]. The normative argument from machine learning makes the cost of hPC's generative assumptions most explicit. Learning by generative "reconstruction" of sensory input allocates a network's capacity towards the subspace explaining most of the input variance, which empirically turns out to be largely uninformative for downstream perceptual tasks [11, 25]. These results suggest that what feedback conveys to early visual cortex is likely not a detailed generative reconstruction of the expected image, but a more general abstract signal about the statistical structure of the current context.

Finally, the hPC postulates about anatomically distinct error and representation neuron populations have also been challenged [30–33]. Systematic literature reviews do not find conclusive evidence for separate error and representation neurons, with positive results often being either equally well explained by feedforward feature detection [30, 31], or largely inconsistent across modalities, tasks and species [32]. Most directly, tracking the same L2/3 V1 PyC somata and their distal apical dendrites across several days in awake mice, shows responses to unexpected pattern-violating stimuli evolve in opposite directions at the soma and dendrites [33]. While the somata integrating bottom-up inputs become less sensitive to pattern-violation, apical dendrites targeted by top-down inputs become more sensitive. Such opposing compartment-specific, experience-dependent plasticity hints at a more complex and dynamic algorithm for predictive learning than one relying on a bank of PPE and NPE units whose functional identity is genetically fixed.

In this work, we take a step back from the pervasive hPC theory, and ask whether visual predictive learning characterized by omission and novelty responses, can be explained by plasticity in L2/3 PyCs following more established connectivity patterns [22, 22, 23]. We first review and analyze recent experimental evidence of experience-dependent plasticity in L2/3 V1 neurons of mice engaging in a visual occlusion paradigm [34], both on population and single-neuron levels. Starting from the experimental data, using a bottom-up modeling approach we design a minimal computational model by implementing different plasticity rules on basal inhibitory, basal excitatory, and apical excitatory synapses in L2/3 PyCs, without relying on feedback-driven inhibition, or separate PPE and NPE neural populations. Our model accounts for the most prominent forms of plasticity observed in these visual occlusion experiments [34], including novelty and omission responses

typically attributed to PPE and NPE neurons. Finally, we compare our model to simple PPE and NPE circuits consistent with the hPC theory, and highlight forms of single-neuron plasticity observed in the experimental visual occlusion dataset which cannot be explained by distinct PPE and NPE neural populations. Altogether, we propose a minimal account of local unsupervised learning of novelty and omission responses through opposing plasticity rules of synapses converging onto different compartments of L2/3 PyCs, consistent with anatomy of L2/3 cortical circuits. How other cortical layers and areas might use these novelty and omission signals to drive representation learning, and other predictive computations, remains a fascinating open question for future work.

## 2 RESULTS

### 2.1 Visual occlusion dissociates sensory and contextual processing in V1

We set out to study predictive learning in V1 neurons, focusing on plasticity of sensory and contextual processing. Decades of research now support the conclusion that activity of neurons in V1 signals much more than sensory presence of low-level features in their classical receptive field (cRF) [3, 35–43]. In addition to their feedforward cRF inputs at the soma and basal dendrites, defining the classical response properties of V1 neurons like orientation tuning; most V1 neurons receive feedback inputs onto their apical dendrites [33, 44–47]. These apical feedback inputs generate extra-classical receptive fields (ecRFs) [39], and depending on the source of their ecRF inputs, activity of simple orientation-selective V1 neurons might be modulated by movement [36, 48, 49], sound [37, 38, 50], or visual context surrounding their cRF [3, 33, 39–43]. Studies using partial visual occlusion of natural scenes provide particularly compelling evidence that human and monkey V1 neurons convey information about specific visual scenes dependent on visual ecRF context even in complete absence of cRF sensory stimulation [3, 40–43]. Thus, visual occlusion represents an ideal setting for studying the differential plasticity of sensory and contextual inputs to V1.

To study experience-dependent long-term plasticity of sensory and contextual processing, we used two-photon calcium imaging to record activity of individual L2/3 PyCs in V1 of awake, running mice during several stages of the visual occlusion paradigm (Fig. 1A). Concretely, we only focused on analysis of neurons with cRFs in the top-left quadrant of the screen, so that this population of neurons would receive no feedforward cRF input and exclusively respond to feedback ecRF input during the occluded (O) trials when the top-left image quadrant was occluded, while receiving both cRF and ecRF input during non-occluded (NO) trials showing full natural images.

At the beginning of the experiment (Naive condition), the mice had no previous exposure to any of the six natural images used in this experiment (Fig. 1B). After recording the brain activity of naive mice in response to passive viewing of both full and occluded versions of all six images, each mouse was familiarized with the same subset of four full images for 2–4 weeks. After familiarization, the brain activity of all mice in response to passive viewing of all the full and occluded versions of all six images was recorded again (Expert condition). Separately analyzing brain activity to the four familiar images shown during familiarization, and the two novel

images only shown in the initial naive and final expert recording sessions, allowed us to quantify omission responses (occlude portion of an expected familiar image) and novelty responses (show an unexpected novel image). Expert mice also participated in a session where only familiar images were shown, paired with water reward to mimic the familiarization setting (Expert Task condition). In this work, we mostly treat Expert and Expert Task conditions interchangeably, but see [34] and our Discussion for more nuance on the differences between effects in Expert and Expert Task conditions.

A complete account of these experiments with methodological and statistical analysis details is presented in previous work [34] - here we aim to synthesize the key findings motivating the development of the circuit model presented in this work.

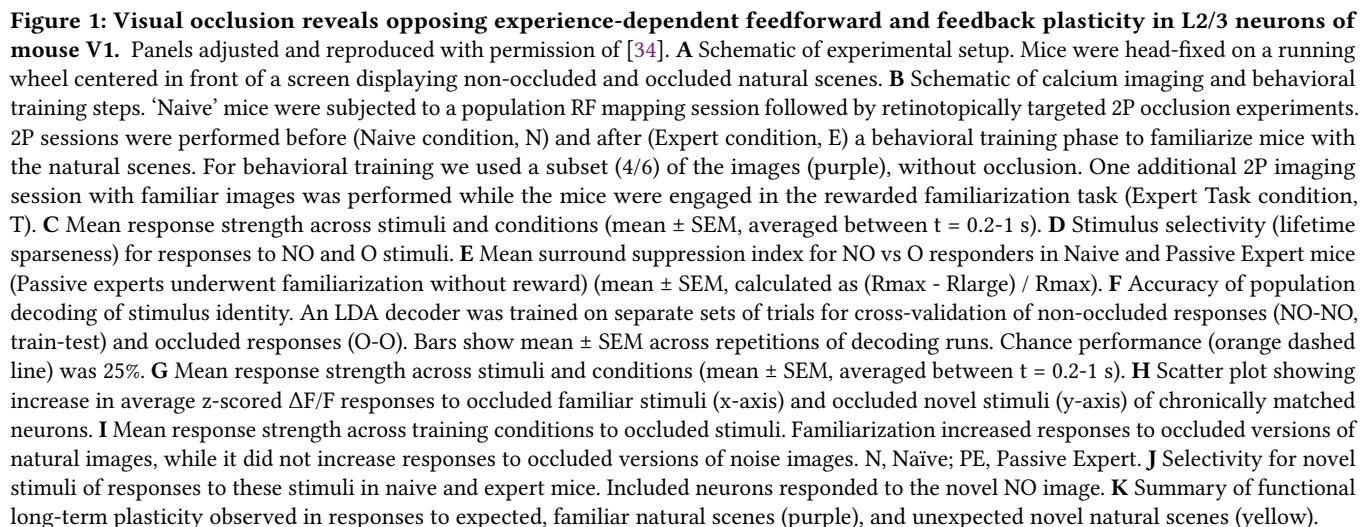
### 2.2 Visual occlusion reveals opposing feedforward and feedback plasticity in V1

We first studied changes to feedforward and contextual processing, related to learned expectations of the four images used during familiarization. Across the population, NO and O responses of individual L2/3 PyCs became progressively decorrelated between Naive, Expert, and Task conditions (Fig. S1A), with a rising separation index reflecting that familiarization drove neurons to become increasingly specialized as either NO or O responders, but rarely both. Consistent with this dissociation, on average NO responses decreased with familiarization while O responses increased in the same population (Fig. 1C). Because O trials by construction contain no cRF input, the emergence of O responses in expert mice implies long-term strengthening of feedback (ecRF) drive. The changes in feedforward cRF and feedback ecRF tuning underlying these opposing changes to NO and O responses respectively were asymmetric: on average NO responses became substantially more selective with familiarization, while O responses remained broadly-tuned independent of experience (Fig. 1D). Follow-up experiments repeating the familiarization paradigm without training water reward (Passive Expert condition), found that neurons responsive to occluded scenes showed greater surround suppression after familiarization (Fig. 1E), suggesting involvement of surround suppression in suppression of sensory responses of omission-signalling expert neurons, and contributing to separation of NO and O responders (Fig. S1A). Consistent with ecRF responses conveying informative feedback signals, scene identity could be linearly decoded from ecRF responses to occluded scenes well above chance, albeit at markedly lower accuracy than from responses to full scenes combining cRF and ecRF input (Fig. 1F).

We next asked whether the same plasticity applies to the two unexpected novel images, which were shown only in the initial Naive and final Expert sessions. In sharp contrast to familiar scene responses, the separation between novel NO and O responders decreased after familiarization (Fig. S1B), while the average responses to unexpected novel scenes increased in both NO and O conditions (Fig. 1G). The change in O responses of individual neurons to novel images was tightly correlated with each neuron’s change in O response to familiar scenes (Fig. 1H,  $R = 0.79$ ), indicating that familiarization-induced changes in ecRF tuning generalized to novel scenes. At the same time, the strengthened ecRF responses were specific to natural scene contexts, reflected by no increase in responses of experts when parts of random noise images were

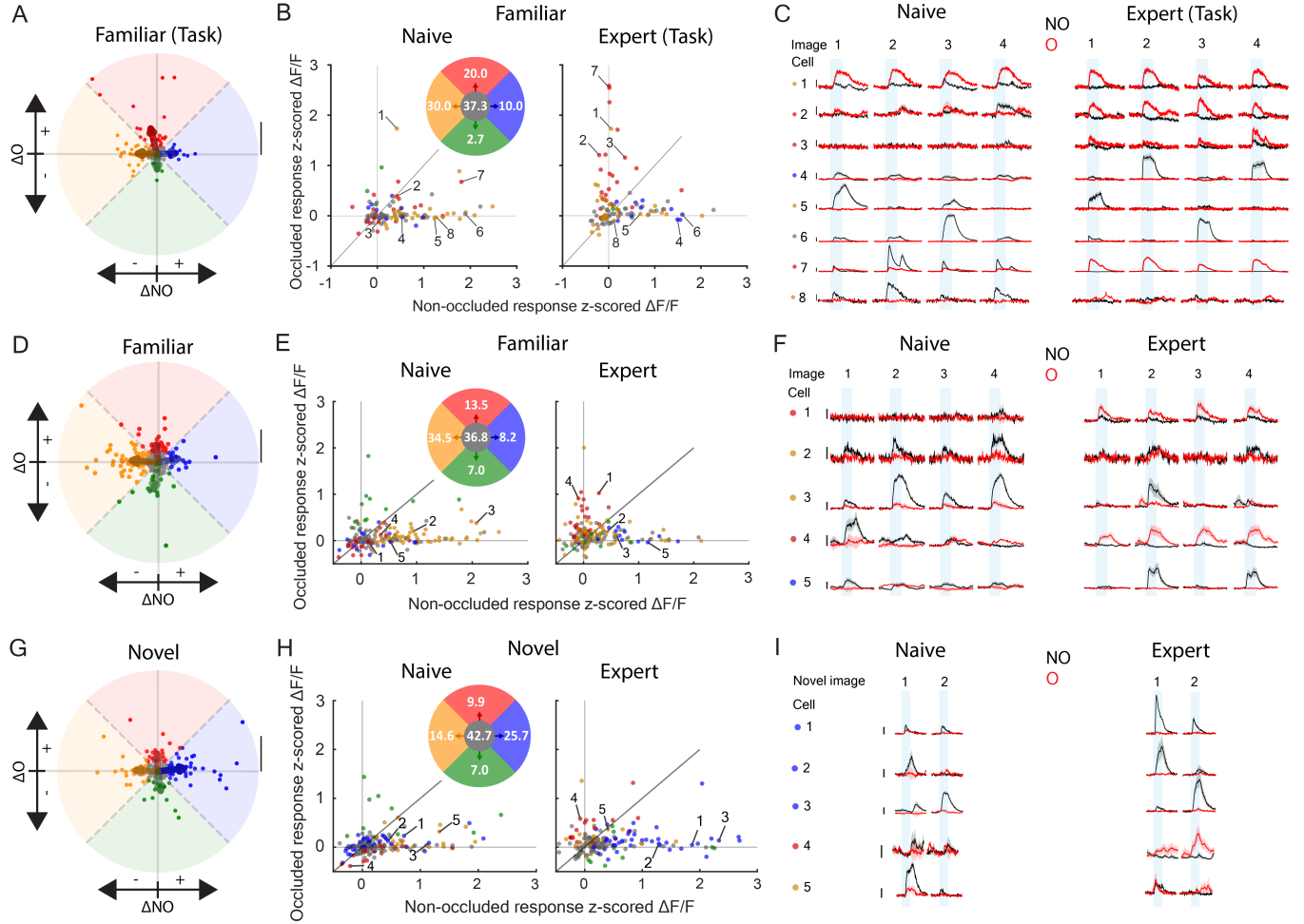


retainment and enhancement of PyCs with narrowly-tuned cRFs, and a potential mechanism for generalized suppression of broadly-tuned cells beyond familiar contexts.



Taken together, these results constrain possible hypotheses about visual predictive learning in L2/3 neurons (Fig. 1K). Learned contextual expectations drive L2/3 in absence of sensory stimulation, and match signatures of predictive omission signals carrying general information about visual scene context. Rather than simply suppressing sensory processing of expected, familiar scenes, L2/3 seems to selectively weaken responses of broadly-tuned neurons, while strengthening responses for highly specific, narrowly-tuned neurons. The tuning effect generalizes to signaling of sensory novelty

responses, which are most prominently enhanced for narrowly-tuned neurons. This suggests that the expectation suppression mechanism most prominently affecting broadly-tuned cells is non-specific and generalizes to unexpected novel stimuli, suggesting lateral coupling to broadly responsive PV interneurons as a possible mechanism. The fact that omission responses also generalize to unexpected novel stimuli, and are less separated from sensory



**Figure 2: Long-term tracking of single-neuron responses reveals principal directions of cRF and ecRF predictive plasticity.** Neurons tracked across naive and expert experimental sessions were classified into stable (gray), increased NO (blue), increased O (red), decreased NO (yellow), and decreased O (green) transition classes based on the angle of their NO and O response change between the naive and expert sessions. **A** Transition plot showing distribution of response changes to familiar images between Naive and Expert (Task) conditions. Colored arrows represent average response change vectors for each transition class. **B** Scatter plots showing average z-scored  $\Delta F/F$  responses ( $t = 0.2-1$  s) to the non-occluded (x-axis) and occluded (y-axis) stimulus of chronically matched neurons in Naive mice and Expert mice performing the task, color coded by the transition class from A. Numbers of the indicated neurons correspond to the example neurons in C. The colored disc displays the percentages of the neurons in each transition class. **C** Neuronal responses of individual, chronically matched example neurons in naive mice (left) and expert mice performing the task (right). Scale bar, 1 z-scored  $\Delta F/F$ . Traces represent mean  $\pm$  SEM over trials. The shaded bars indicate stimulus epochs. **D** Same as A but for unrewarded Expert condition. **E** Same as B but for unrewarded Expert condition. **F** Same as C but for unrewarded Expert condition. **G** Transition plot showing distribution of response changes to novel images between Naive and Expert conditions. **H** Same as E but for novel images. **I** Same as F but for novel images.

responses compared to familiar stimuli, indicates that perhaps familiarization generally strengthens ecRFs in a way which also contributes to the strengthened novelty responses of PyCs selectively tuned to cRF features in novel scenes.

To obtain further insight on directions of single-neuron cRF and ecRF plasticity, we extended the single-neuron analyses of previous work analyzing the same visual occlusion experiments [34]. Concretely, we asked in what directions do sensory and contextual responses of individual L2/3 neurons evolve with familiarization (Fig. 2). The most prominent single-neuron transition class was a reduction of sensory responses to NO familiar stimuli, prominently visible in both Expert (Task) (Fig. 2A-C, yellow cells, examples 1, 5, 8) and Expert (Fig. 2D-F, examples 2,3) conditions. Second most common were transitions towards increased contextual omission responses to O familiar stimuli, clearest in the rewarded Expert (Task) condition (Fig. 2A-C, red cells, examples 2, 3), but also observed in the unrewarded Expert condition (Fig. 2D-F, example 1). These transitions alone would be largely compatible with hPC [14], corresponding to PPE plasticity reflected in suppressed sensory responses to NO expected stimuli, and NPE plasticity reflected in enhanced top-down predictions triggering omission responses to occluded familiar stimuli.

However, several single-neuron transitions we observed suggest a departure from classical hPC. First, we observed L2/3 PyCs which transitioned from predominant NO responders into predominant O responders (Fig. 2A-C example 7; D-F example 4). This mode of plasticity can be accommodated neither by PPE cells which according to hPC theory [17] do not receive direct ecRF inputs and cannot show omission responses; nor NPE cells which do not receive direct cRF inputs and cannot respond to NO scenes without also responding to their O counterparts. Second, a subset of L2/3 PyCs strengthened their NO responses to the expected, familiar stimuli (Fig. 2A-C blue cells, example 4; D-F example 5), which defies the normative principle of hPC that L2/3 neurons should only signal differences between predictions and sensory evidence [14]. Third, many individual L2/3 neurons also strongly changed their response patterns to unexpected, novel images (Fig. 2G-I). This suggests that the learning mechanisms underlying predictive enhancement of contextual signals and suppression of expected sensory inputs are not specific to the low-level features of familiar stimuli in a generative hPC sense, but instead capture general statistics of natural images which can generalize to novel scenes.

### 2.3 Bottom-up circuit modeling of predictive learning in L2/3 neurons

These observations motivated mathematical formalization of a minimal circuit model that could explain these effects in terms of experience-dependent bottom-up cRF and top-down ecRF plasticity of individual L2/3 PyCs. The circuit connectivity was directly constrained by three aspects of L2/3 function and anatomy in V1:

- (1) L2/3 PyCs should generally receive both feedforward cRF and feedback ecRF inputs [23, 33, 39], and experience-dependent plasticity should determine synaptic strength of those inputs.
- (2) Suppression of predictable responses in L2/3 should be a function of widespread feedforward-lateral adaptation and inhibition, instead of sparse top-down inhibition [22–24].

- (3) L2/3 PyCs should be modeled as two-compartment neurons where basal-somatic compartments integrate cRF and lateral input, and apical compartments integrate ecRF input [44–46], with compartment-specific plasticity rules [33].

Based on these constraints, we naturally arrived at a minimal L2/3 circuit architecture where the basal compartment of each PyC receives feedforward excitatory cRF inputs and lateral inhibitory inputs, while the apical PyC compartment receives top-down excitatory ecRF (Fig. 3A, detailed treatment in Methods). In addition to the PyC, each circuit contained a parvalbumin interneuron (PV), which received feedforward and lateral excitatory inputs, thereby providing feedforward and lateral surround suppression to the PyC, without directly communicating top-down predictions. In this work, we were interested in studying plasticity rules underlying predictive learning in individual L2/3 PyCs, (Fig. 1 & 2) - and thus focused on the simplest circuits with 1 PyC and 1 PV, only studying plasticity of PyC synapses. Nevertheless, our implementation permits a more general study of L2/3 networks consisting of interconnected PyC and PV populations.

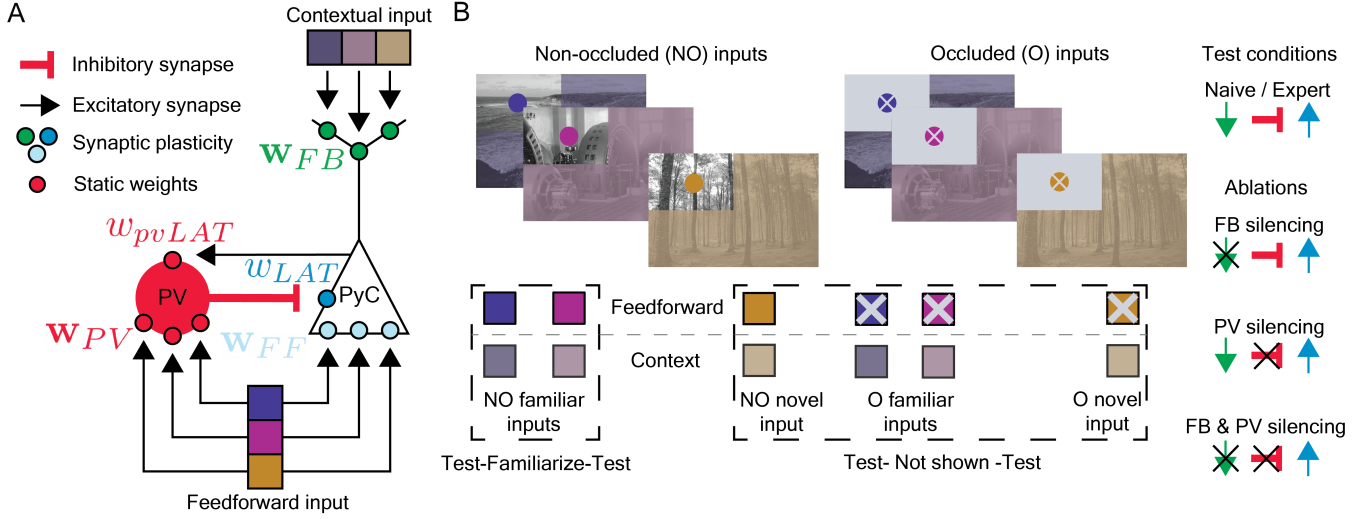
To derive the plasticity rules at the feedforward cRF, lateral inhibitory, and feedback ecRF synapses of the PyCs, we turned to the results from the visual occlusion experiments (Figs. 1 & 2). The dominant single-neuron transition with familiarization was a reduction of NO responses to familiar images (Fig. 2A-F, yellow class), reflected in the average drop in NO responses at the population level (Fig. 1C). The simplest local rule that produces such input- and cell-specific long-term suppression is an anti-Hebbian depression of the corresponding basal  $w_{FF}$  synapses as a function of postsynaptic PyC firing and presynaptic cRF stimulation.

$$\Delta w_{FF}(t) \propto -\text{cRF}(t) \cdot \text{PyC}(t) \quad (1)$$

Such a rule is also grounded in documented subcellular mechanisms: sustained and repeated postsynaptic activity is well known to recruit multiple pathways whose net effect is a long-term reduction of subsequent feedforward drive, including slow homeostatic downregulation of excitability through AMPA-conductance scaling [51, 52], and activity-dependent plasticity of the axon-initial-segment [53, 54].

Beyond the stimulus-specific loss of NO responses to familiar images, familiarization produced generalized expectation suppression effects that could not be attributed to depression of individual cRF synapses alone: large proportion of PyCs showed reduced NO responses to novel unexpected scenes (Fig. 2G-I), while the NO responder population became substantially more selective to both familiar and novel scenes (Fig. 1D, J). The generalization of these effects to novel scenes never presented during familiarization requires a plasticity mechanism that must be non-selective at the input level while still preferentially suppressing broadly-tuned cells. The increased surround suppression in expert O responders (Fig. 1E) hints at one such mechanism compatible with our L2/3 circuit. Namely, Hebbian potentiation of the lateral inhibitory synapse  $w_{LAT}$  (converging on the PyC soma) as a function of postsynaptic PyC firing and presynaptic PV firing implements exactly the desired generalized expectation suppression in our simple circuit model.

$$\Delta w_{LAT}(t) \propto \text{PV}(t) \cdot \text{PyC}(t) \quad (2)$$



**Figure 3: Evidence-based design of a minimal L2/3 circuit model for predictive learning.** **A** Minimal circuit model schematic. PyC receives basal excitatory feedforward input ( $w_{FF}$ ), and apical excitatory contextual feedback ( $w_{FB}$ ). PV receives excitatory feedforward input ( $w_{PV}$ ) and excitatory lateral input from the PyC ( $w_{pvLAT}$ ). Feedforward inhibition from PV acts as divisive normalization on PyC somatic activity (via  $w_{LAT}$ ) which integrates basal and apical inputs. Only PyC synapses undergo plasticity. **B** Simplified simulation of the visual occlusion paradigm for testing cRF and ecRF plasticity in minimal circuit models of L2/3 PyCs. The simulation mirrors a simplified version of the real experimental setting (Fig. 1A,B) by using two familiar inputs (1, 2) and one novel input (3). Each non-occluded (NO) image is represented as a combination of feedforward cRF and contextual ecRF inputs. We simulate occluded (O) images by zeroing-out the feedforward input for each image. Similar to the experiment, we perform naive testing on NO and O versions all inputs, but only familiarize with NO familiar inputs. After familiarization we perform expert testing on NO and O versions all inputs; also coupled with ablations silencing PV inhibition and contextual feedback. Plasticity is only active during the familiarization phase.

Compared to selective, narrowly-tuned PyCs, broadly-tuned PyCs respond to more stimuli on average and thus co-activate more frequently with the broadly-tuned PVs, making their  $w_{LAT}$  potentiate faster compared to narrowly-tuned PyCs. The resulting expectation-suppressing feedforward inhibition affects the responses of PyCs across familiar and novel contexts. This rule mirrors two established results: cortical inhibitory circuits are homeostatically wired so that the most active PyCs recruit proportionally more inhibition [55]; and Hebbian potentiation of inhibitory synapses in a feedforward-inhibited circuit has been shown to generate novelty- and adaptation-like response profiles closely matching those otherwise attributed to prediction-error neurons [56].

Finally, the visual occlusion experiments constrained possible apical feedback plasticity rules along two axes. First, O responses to familiar images increase with experience (Fig. 1C) and a many initially unresponsive PyCs transition into predominant O responders with familiarization (Fig. 2A-C example 2; D-F example 1), requiring plasticity to potentiate ecRFs of cells whose post-synaptic firing is low. Second, the change in O responses to familiar and unexpected novel images is tightly correlated (Fig. 1H), suggesting that ecRF plasticity must generalize across contexts, while maintaining a degree of specificity to enable linear discriminability of familiar contexts from population responses to ecRF stimulation alone (Fig. 1F). These two constraints together motivate an apical learning rule that broadly potentiates  $w_{FB}$  synapses as a function of presynaptic ecRF context in an inverse relationship with PyC firing.

$$\Delta w_{FB} \propto \frac{S^c \cdot \text{ecRF}(t)}{\text{PyC}(t) + 1} \quad (3)$$

The inverse dependence on postsynaptic PyC firing ensures that ecRF strengthening is largest in cells whose sensory drive is weakest, enabling silent neurons to reliably transition into signaling omissions after familiarization. The context-specificity matrix  $S^c$  lets familiarization-driven potentiation of ecRF synapses generalize across contexts (thanks to nonzero off-diagonal elements) while retaining enough context-selectivity to support discrimination of familiar scene identity from ecRF responses alone (diagonal elements still dominate). While this learning rule remains somewhat speculative, the proposed inverse dependence of apical plasticity on PyC activity could be implemented by subcellular mechanisms in the apical compartments, which receive information about somatic activity via backpropagating action potentials [46, 57]. The generalized potentiation of L2/3 ecRFs via  $S^c$  is consistent with the growing sentiment that predictive learning and computation happen at a higher and more general level of abstraction [2, 11, 24–29].

When developing the model, we encountered an apparent contradiction between our results and the literature. On one hand, canonical results on apical dendrite physiology hold that top-down feedback stimulating apical dendrites mostly acts like multiplicative gain on somatic activity [44]. On the other hand, our occlusion experiments clearly show that ecRF stimulation alone can drive PyC activity in absence of direct somatic or basal cRF stimulation (Figs. 1 & 2). To address this issue, we decomposed the apical



contribution to PyC firing into two functionally distinct terms: a low-threshold multiplicative gain  $\gamma$ , and a high-threshold additive drive  $\beta$  (Eq. 6, Methods). This allows us to observe the canonical multiplicative apical amplification of somatic activity [44] across a broad range of apical activity levels; while requiring strong apical activation to directly drive the PyC firing. Such distinction is consistent with the observation that suprathreshold apical drive requires local NMDA plateaus [45], potentially in coincidence with back-propagating action potentials and BAC firing [57]. Multiplicative apical amplification also allows us to explain one of the perplexing transitions observed in our data contributing to the increased separation of familiar NO and O responders - namely the subset of neurons whose NO responses strengthen with familiarization without strengthening their O responses (Fig. 2C example 4; F example 5). With multiplicative gain, these transitions can still be explained by potentiation of ecRF synapses resulting in increased gain on NO responses in neurons where that potentiation does not cross the apical drive threshold, and feedforward adaptation mechanisms preserve some cRF drive. A single feedback rule for ecRF potentiation can then generate both apically-amplified +NO responses in cells retaining some cRF drive, and pure +O omission responses in more completely adapted cells, depending on the initial cRF and ecRF tuning.

A separate concern arose from a naive subtractive implementation of feedforward inhibition, which naturally drove the activity broadly-tuned cells significantly below baseline NO firing with familiarization. We did not observe this trend in the data (Figs. 2B,E; S1), where even cells undergoing drastic feedforward adaptation or suppression still showed baseline firing rates in Expert NO conditions. Reformulating PV inhibition as divisive normalization [58] of somatic activity (Eq. 4) avoided this failure mode, and mirrored recent predictive coding work explicitly modeling cortical PV interneurons as divisive [59].

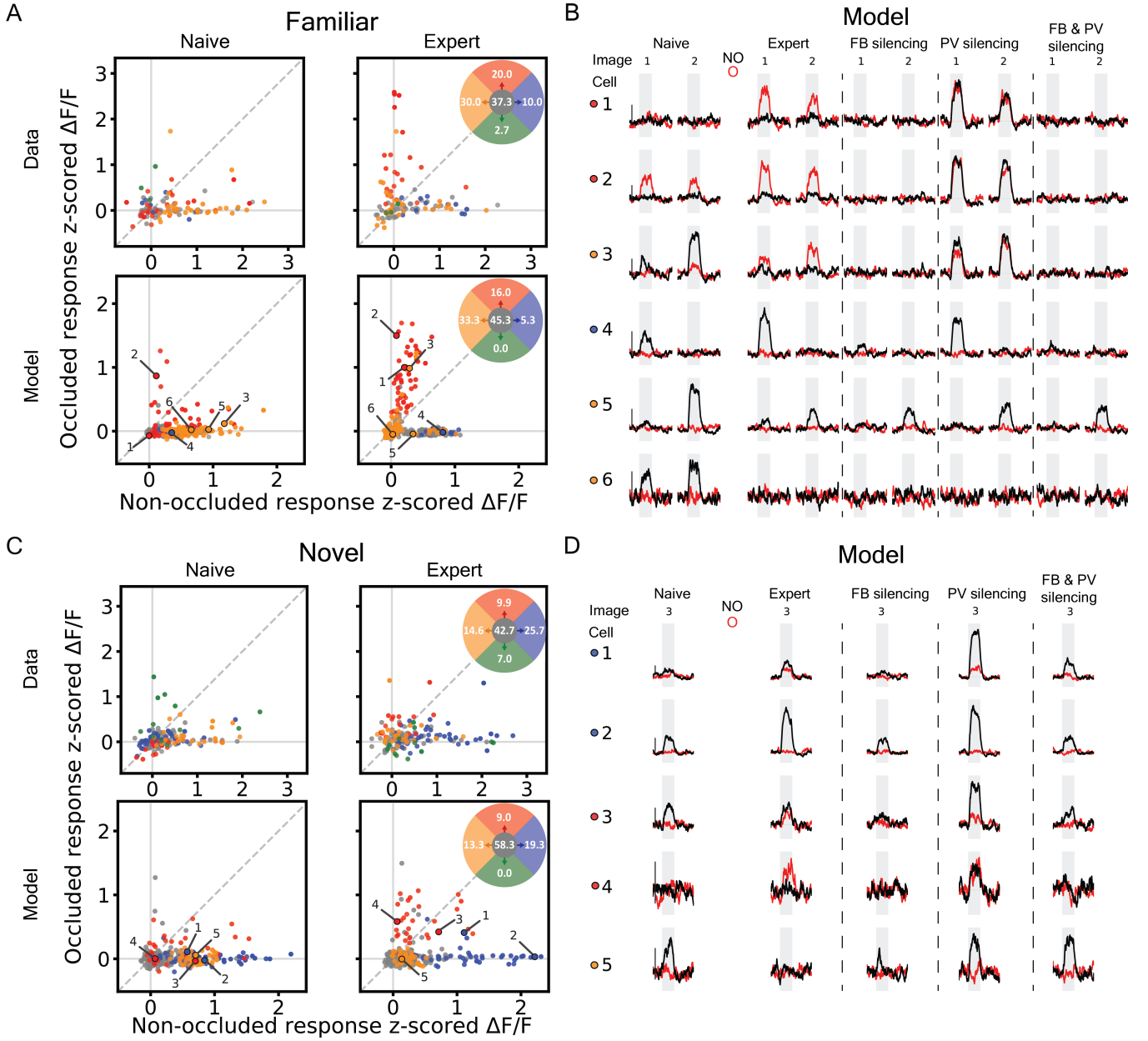
Finally, we wanted to make the distinction between broadly and narrowly-tuned cells more explicit. Because both the experiment and our simulations used ITIs long enough for previous trial-evoked firing to die out, initial versions of our model had no mechanism by which consistently repeated firing in previous trials could directly influence firing or plasticity in future trials, other than through long-term feedforward depression. In contrast, cortical PyCs are known to exhibit spike-frequency adaptation on multiple timescales, from fast  $\text{Ca}^{2+}$ -activated potassium currents that relax within tens of milliseconds [60, 61], to slower short-term depression components that recover over seconds [62, 63], and other feedforward adaptation mechanisms that are slower still [51–54]. To reflect these various timescales of firing adaptation in cortical PyCs, we introduced two adaptation variables - a fast  $a_f$  responsible for within-trial adaptation to sustained PyC firing; and a slow  $a_s$  accumulating over several trials that modulates the rate of feedforward depression (Eq. 5). The consequence of direct gating of cRF plasticity by  $a_s$  is that cRF depression is the strongest in broadly-tuned cells that activate across many subsequent trials, while narrowly-tuned cells activate more rarely and experience less cRF depression. This effect further compounds the suppression of broadly-tuned cells produced by  $w_{\text{LAT}}$  potentiation, while simultaneously increasing the probability of narrowly-tuned cells to preserve larger portion of their cRF

tuning. Combined with familiarization-strengthened apical gain, such cells were expected to explain the +NO transition class.

## 2.4 Circuit model with opposing plasticity rules produces novelty and omission responses

To probe these plasticity rules on the same task as the experimental data, we simulated a simplified version of the visual occlusion paradigm using two familiar images and one novel image (Fig. 3B). Each non-occluded (NO) image was represented as a pair of one-hot feedforward cRF and contextual ecRF inputs, and the corresponding occluded (O) image was obtained by zeroing the cRF input while leaving the ecRF input intact. We first probed naive circuits on all six stimulus conditions ( $\text{NO/O} \times 3$  images), then familiarized neurons only with the two familiar NO images, and re-probed expert circuits on all six conditions. At the expert testing stage, each condition was repeated under three additional ablation regimes obtained by silencing the PV inhibition, silencing the ecRF feedback, and silencing both to isolate pure cRF responsiveness. Plasticity was active only during familiarization, so that all changes between naive and expert responses reflected plasticity during the familiarization phase. To generate a diversity of naive sensory and contextual responses comparable to the experimental data, we randomly varied the initial conditions (including cRF and ecRF tuning amongst other parameters) of our circuit models around transition templates designed to capture key single-neuron transitions observed in the data (for simulation details see Methods).

Our population of simulated L2/3 PyC circuits, sampled from the transition templates and probed on a simplified version of the same visual occlusion protocol as the experiment, qualitatively reproduced the population-level statistics and all the dominant single-neuron transition directions observed in the data (Fig. 4A,C), both to familiar and novel images. The most frequent transition class for familiar images was the reduction of sensory NO responses (Fig. 4A, yellow; examples 5–6 in Fig. 4B), matching the dominant –NO class in the data (Fig. 2A–F). In the model, cells with broad cRF tuning for both familiar images fire repeatedly during familiarization, driving strong accumulation of the slow adaptation variable  $a_s$  which dynamically facilitates rapid anti-Hebbian depression of their  $w_{\text{FF}}$  synapses. This mechanism is clearly visible from isolation of cRF tuning by simultaneous FB & PV silencing (Fig. 4B), where familiarization completely adapts pure cRF responses in broadly-tuned cells (examples 3, 6), while cRF responses of narrow cells are reduced, but remain nonzero (examples 4, 5). Like in the data, the second dominant transition direction was an increase in omission responses to O familiar images (Fig. 4A, red; examples 1–3 in Fig. 4B). Our model allowed several trajectories to developing omission responses (Fig. 4B), including starting from silent (example 1), already O responsive (example 2), or NO responsive (example 3) naive states. Regardless of naive state, omission-signaling expert neurons shared common signatures of an adapted-away cRF drive (revealed by simultaneous FB & PV silencing), and prominent feedforward suppression (revealed by a surge of ecRF-driven responses with PV silencing), matching experimental data (Fig. 1E). A small subset of simulated cells strengthened NO responses to familiar images (Fig. 4A, blue; example 4 in Fig. 4B): a transition that is hard to reconcile with hPC theory. In our model, this transition can only occur in



**Figure 4: A minimal L2/3 circuit reproduces predictive learning of novelty and omission responses without prediction-errors.** **A** Responses to familiar images before (Naive) and after familiarization (Expert), for real data (Expert Task condition) and model neurons. Neurons are colored by the dominant transition direction of their familiarization-induced response change to NO and O images (see Fig. 2). Model responses ( $n=300$ ) are z-scored based on the naive pre-stimulus period ( $t = 0-1s$ ). **B** Example responses of individual neurons to familiar images, before and after familiarization. Includes post-familiarization perturbation experiments with FB silencing, PV silencing, and simultaneous FB & PV silencing. Numbered examples are highlighted in **A**. Scale bar indicates 1 SD. **C** As in **A**, but for the novel image. Data are shown for the same PyCs whose average responses to familiar images are plotted in **A**. **D** Same as **B** but for novel image. Highlighted PyCs in **C-D** are distinct from those in **A-B**.

narrowly-tuned cells, with two key properties: their narrow feed-forward tuning limits  $a_s$  accumulation and therefore protects  $w_{FF}$  from complete depression, and their nonzero apical feedback is

potentiated to a level that engages the low-threshold apical gain  $\gamma$  but stays below the high-threshold apical drive  $\theta_d$ . Such configurations give rise to multiplicative apical amplification of a partially

preserved basal response. FB silencing confirms this mechanism: removing FB removes the gain and only cRF responses adapted relative to the naive state remain. This transition direction empirically justifies the distinction between multiplicative apical gain from purely additive apical drive, and dynamic firing-frequency dependent feedforward adaptation.

Extending the analysis to the untrained novel image (Fig. 4C,D) shows that our models capture the generalizing effects of familiarization observed in the data. Cells that show novelty responses to novel images (increased NO responses), without showing omission responses (Fig. 4D, example 2) are narrowly-tuned PyCs preferentially tuned to the novel input. These novelty responses come purely from the generalizing effects of feedback plasticity resulting in apical amplification of cRF responses. Isolating cRF responses with FB silencing confirms cRFs of these cells are not directly affected by familiarization. The high specificity of these cells to novel images is also confirmed by complete lack of effect of PV silencing on these novelty responses - indicating these cells were not active during familiarization and did not recruit any PV inhibition. Cells that reduce their responses to novel NO images (yellow) are broadly-tuned cells that acquired strong  $w_{LAT}$  through repeated co-firing with the adjacent PV during familiarization: because the potentiated inhibition is triggered by the shared PV rather than by image-specific FF drive, it acts equally on responses to unexpected novel inputs. PV silencing confirms this mechanism, showing unaffected cRF drive in absence of generalized strengthened inhibition (Fig. 4D, cell 5). Finally, a subset of cells produce novel O responses (red, Fig. 4D, examples 1,3,4) purely from cross-context  $w_{FB}$  potentiation crossing  $\theta_d$  for the novel context. FB silencing removes these responses entirely, confirming that novel omission responses inherit their ecRF drive from the same generalized apical mechanism that produces the familiar omission responses, consistent with the tight cross-image correlation of  $\Delta O$  observed in the data (Fig. 1H). Because no image-specific cRF anti-Hebbian depression occurred for the novel image, cRF and ecRF responses of cells signaling omissions are not decorrelated, which is why experts show more simultaneous NO and O responses to novel images (Fig. 4D, example 1) compared to familiar images (Fig. S1B).

Together, these simulations show that opposing plasticity mechanisms within individual PyCs, in combination with feedforward-driven inhibition, could in principle account for the concurrent reduction in responses to familiar stimuli and enhancement of responses to novel and occluded stimuli across non-overlapping PyC populations. Notably, all of these transitions could be captured by plasticity occurring at the level of individual PyCs only.

## 2.5 Comparison with prediction-error circuits

To make the theoretical contribution of our context-contrasting (CC) circuit model (Fig. 5A) more explicit, we compared it against the two canonical L2/3 prediction-error motifs from the hPC literature, namely the positive prediction-error (PPE) and the negative prediction-error (NPE) circuits [18–20, 59, 64]. To isolate the role of L2/3 PyCs, which are the focus of this work, we studied simplified versions of these prediction-error circuits (Fig. 5B,C), not including L5, multiple interneuron types, and a hierarchy of interconnected cortical regions often included when modeling prediction error circuits [59, 64]. Still, our simplified implementations

of these circuits capture the specific role that hPC theory assigns to L2/3 PyCs, and thus provide a fair reference for our question of how far L2/3 PyC plasticity alone can go in reproducing the observed response transitions.

In a PPE circuit, the PyC receives direct sensory cRF excitation and subtractive inhibition from a PV cell driven by top-down context; its firing represents the portion of sensory evidence not explained by prediction. These responses can be interpreted as signaling novel sensory evidence that has yet not been taken into account by the internal generative model. In an NPE circuit the roles are swapped: the PyC receives direct contextual ecRF excitation and subtractive inhibition from a PV cell driven by sensory input, so its firing represents the portion of the prediction unmatched by sensory evidence. These responses can be interpreted as signaling omissions of sensory evidence expected by the internal generative model. In both circuits, prediction errors can be minimized either by Hebbian potentiation of the inhibitory PV→PyC synapse, or by anti-Hebbian depression of the excitatory PyC synapses (Fig. S2). In line with our focus on experience-dependent plasticity of cRF and ecRF responses of L2/3 neurons, we focused on the prediction-error circuits implementing anti-Hebbian depression of their excitatory synapses. Because the voltage of each PyC computes the signed difference of a single excitatory and a single subtractive term, the sign of both plasticity rules flips naturally at the moment this difference crosses zero. This is a powerful homeostatic mechanism which guards against runaway or insufficient plasticity, that our CC circuit does not have.

Simulating matched CC, PPE and NPE populations on the same simplified occlusion protocol from Fig. 3, highlights the modes of plasticity that each circuit motif can and cannot produce (Fig. 6). PPE circuits can capture the –NO and (through sign-flipping of anti-Hebbian plasticity) +NO transition classes for familiar images, but by construction cannot produce any O response as they lack excitatory cRF inputs (Fig. 6B,F). Furthermore, the +NO PPE transitions occurred only in PyCs initially suppressed below baseline firing. After familiarization their cRF responses strengthened to cancel out with top-down inhibition during NO trials. This is unlike the +NO transition observed in the data, characterized by strengthened above-baseline NO responses after familiarization. NPE circuits complement this by generating familiar +O, –O and –NO transition classes, but their NO and O response changes remain tightly correlated, as both types of responses can only come from top-down ecRF input, as these cells lack any bottom-up excitation (Fig. 6C,G). Furthermore, both PPE and NPE circuits produce no plasticity of responses to novel images, and retain their initial tuning to unexperienced stimuli because their plasticity is specific to synapses stimulated during familiarization (Fig. 6B,C). Consequently, neither PE motif reproduces the coexistence of –NO and +O plasticity within a single neuron, nor the empirical generalization of expectation suppression and omission signaling to novel images (Fig. 6D,H).

In contrast, the CC circuit, matches most transition classes for both familiar and novel images (Fig. 6A,E), and it also reproduces the transitions involving simultaneous plasticity of NO and O responses that the PE circuits cannot capture (Fig. S3).

Two structural properties of the CC circuit are responsible for this expressive gap. First, apical and basal plasticity are independent,

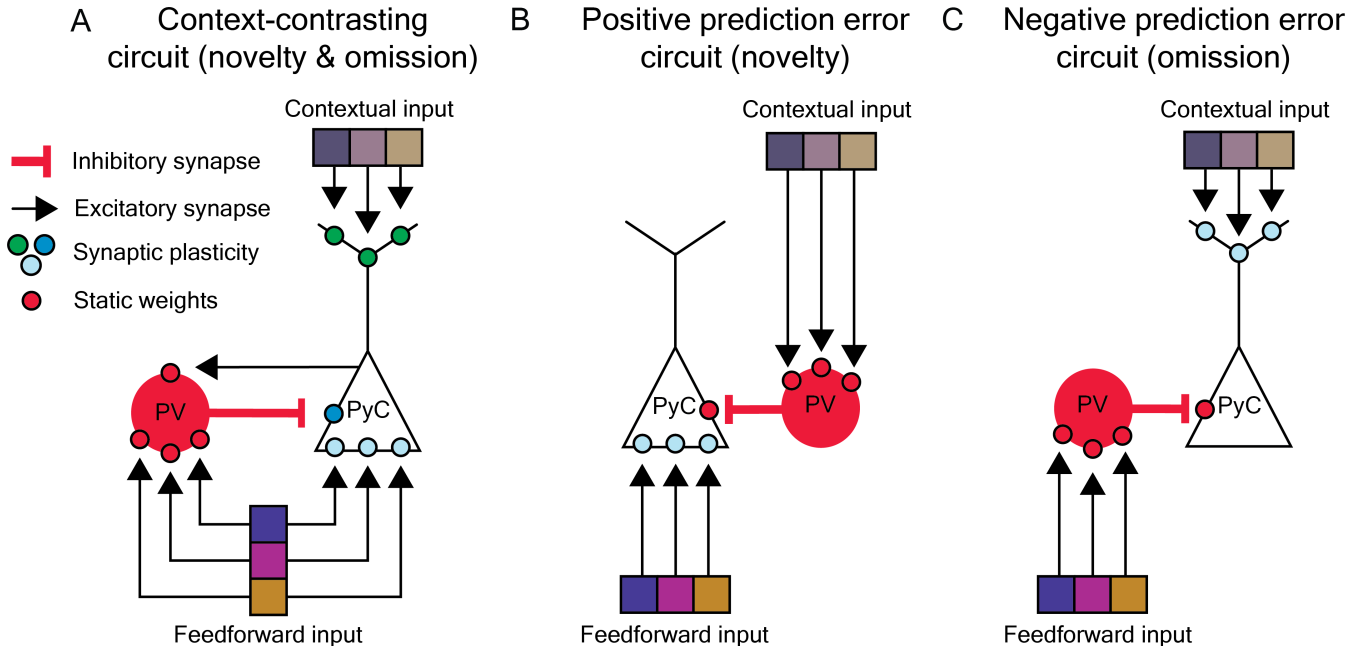
so a single PyC can simultaneously learn to strengthen ecRF drive and depress cRF drive without those changes being driven in the same direction by a single signed error. Second, the generalization channels built into the CC plasticity rules (off-diagonal entries of  $S^c$  for feedback and shared-PV coupling for lateral inhibition), allow familiarization to shape responses to novel stimuli the network has never been exposed to, which is exactly the property our simple PE circuits lack.

### 3 DISCUSSION

In this work we asked how far plasticity in individual L2/3 PyCs can go in explaining visual predictive learning, without top-down inhibition or genetically distinct prediction-error populations. Extending the single-neuron analyses of the visual occlusion dataset [34], we found that sensory and contextual responses evolve along a few directions with familiarization (Fig. 2), three of which are difficult to reconcile with PPE and NPE units of fixed identity: sensory responders converting into contextual responders, strengthening of sensory responses to expected familiar images, and visual experience with a small set of familiar natural scenes inducing plasticity generalized to unexpected novel scenes. Guided by these transitions and by current knowledge about the anatomy of L2/3 circuits in V1 [22, 23, 33, 46], we propose a minimal account of local unsupervised learning of novelty and omission responses in L2/3 PyCs receiving feedforward and lateral PV inhibition (Fig. 3). We suggest three local plasticity rules act in opposition inside individual PyC neurons, and drive predictive learning: anti-Hebbian depression of

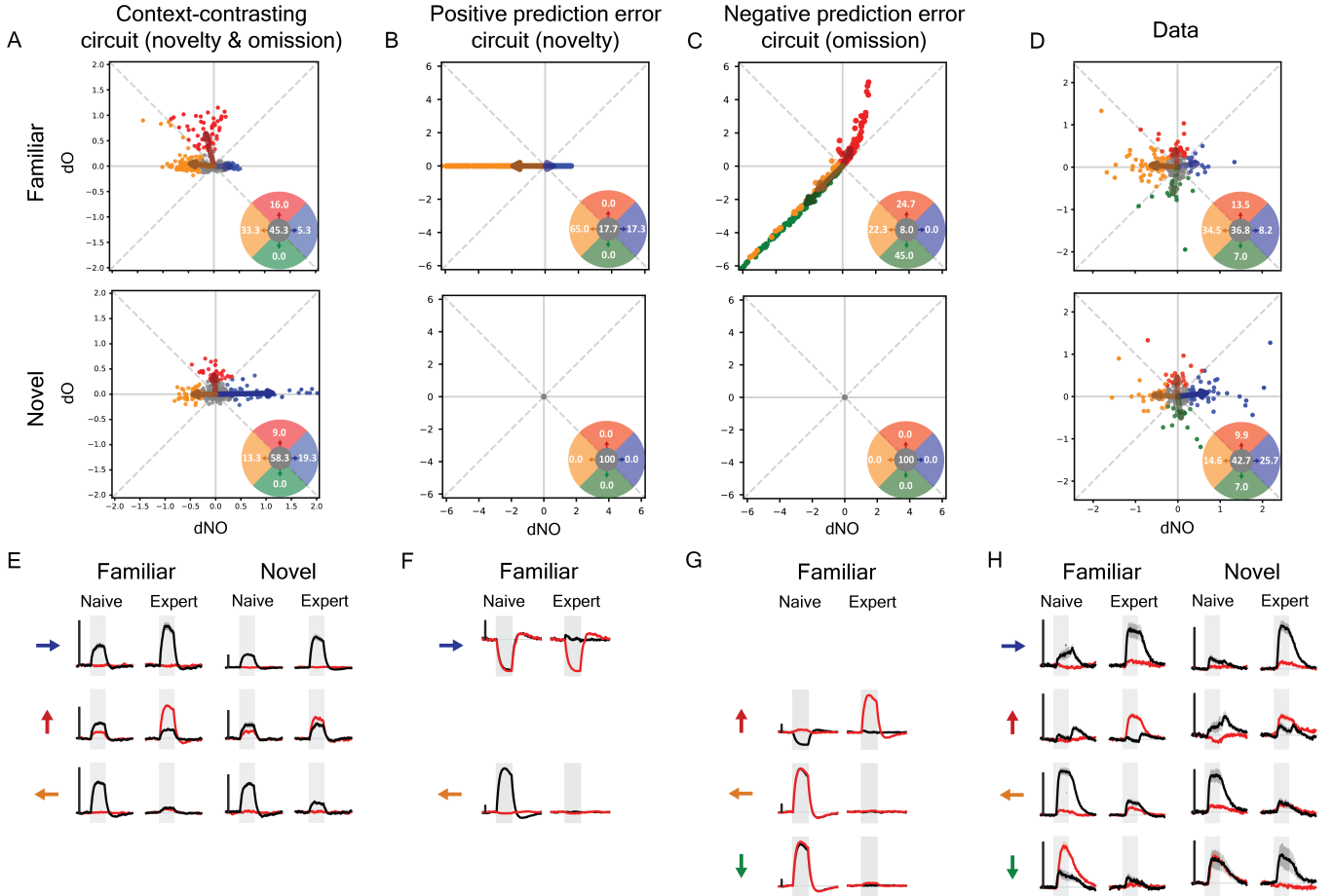
basal cRF synapses gated by a slowly accumulating firing-frequency adaptation variable; Hebbian potentiation of divisive lateral inhibition; and broad potentiation of apical ecRF synapses generalizing across contexts. Simulated on a simplified visual occlusion protocol, this circuit reproduced the population statistics and all dominant transition classes, for familiar and novel images alike (Fig. 4).

Comparison with simplified prediction error circuits, capturing L2/3 connectivity patterns assumed by hierarchical predictive coding [17, 59], revealed a number of transitions observed in the data that our CC circuit was better suited to reproduce. Together, the PPE and NPE circuits together could explain weakening of sensory responses (-NO), and strengthening of contextual responses to familiar images (+O), and even partially captured the rare weakening of contextual responses (-O) which our CC model cannot explain (Fig. 6). However these prediction error circuits were structurally unable to reproduce the strengthening of familiar sensory responses (+NO) observed in the data, nor the generalized effects of experience-dependent plasticity observed in response transitions to novel images. Through silencing experiments (Fig. 4), we could locate properties of the CC circuit underlying these various forms of plasticity not observed in prediction error circuits. FB silencing reduced sensory responses of +NO cells to an adapted cRF response for familiar stimuli, and an unchanged cRF response for novel stimuli; while abolishing increased novel contextual responses entirely - showing that both transitions arise from the same generalized apical potentiation. PV silencing left the cRF drive of novel -NO cells intact, locating their suppression in lateral inhibition shared



**Figure 5: L2/3 Circuit models for predictive learning of novelty and omission signals.** **A** Context-contrasting PyCs follow opposing plasticity rules in apical and basal compartments, capable of producing both novelty and omission responses with experience. **B** Positive prediction error PyCs compute how much of sensory evidence is not explained by top-down prediction, and minimize this error via anti-Hebbian plasticity of their cRF feedforward synapses. **C** Negative prediction error PyCs compute how much top-down prediction exceeds sensory evidence, and minimize this error via anti-Hebbian plasticity of their ecRF feedback synapses.





**Figure 6: Comparison of principal experience-dependent plasticity motifs across L2/3 circuit models.** A-D Familiarization-induced response-change ( $\Delta NO$ ,  $\Delta O$ ) distributions for populations of CC (A), PPE (B), NPE (C), and empirically recorded (D) L2/3 PyCs, split by familiar (top row) and novel (bottom row) images. Each point is one cell; colors indicate transition class as in Fig. 2A (blue: +NO; red: +O; yellow: -NO; green: -O; gray: stable). Inset discs report the percentage of cells in each transition class. E-H NO (black) and O (red) responses averaged across all cells from each observed transition class (indicated by the colored arrows) in Naive and Expert conditions, for each of the four populations of L2/3 PyCs in A-D.

across images rather than in image-specific feedforward depression. In expert omission responders it unmasked a surge of ecRF-driven activity, indicating that the population-level separation of sensory and contextual responders rests on feedforward and laterally-driven suppression via broadly-tuned interneurons.

The process of designing our L2/3 circuit yielded four insights, all reflected in the CC circuit presented in this work. First, plasticity confined to the synapses of a single PyC is sufficient to explain predictive learning of novelty and omission responses, as every transition class was captured without plasticity of synapses on the PV interneuron. In initial stages we considered circuits with plasticity on all synapses, but finding multiple independent mechanisms for adaptation of sensory responses to be redundant, we chose to focus on PyC plasticity only. Second, separating the apical contribution into a low-threshold multiplicative gain and a high-threshold additive drive helped us reconcile canonical apical amplification effects [44] with the ability of ecRF input alone to drive firing

during occlusion, letting one feedback rule generate both apically-amplified +NO responses and pure +O responses. Initial versions of our model considered only additive apical drive, which could not reconcile these effects. Third, PV inhibition had to be divisive rather than subtractive [58], since our initial subtractive implementation drove broadly-tuned cells well below their naive baseline, which was not seen in the data. Lastly, gating cRF depression with an adaptation variable accumulating over the longer timescale of several trials makes plasticity dynamically depend on tuning breadth without explicit selectivity-based parameters, since broadly-tuned cells activate on more trials and thus adapt their cRFs faster.

### 3.1 A predictive coding perspective

Despite the fact that the CC circuit was designed to explain plasticity observed during the visual occlusion paradigm [34], several effects observed in this work nevertheless remain open to a predictive

coding interpretation. For one, many chronically tracked neurons retained their naive response pattern to novel stimuli (Fig. 2), consistent with PE circuits, which show no plasticity of responses to stimuli whose synapses were not driven during familiarization (Fig. 6), at least without learned confidence or attention signals. Further analysis of novel scene plasticity induced by different kinds of familiar scene plasticity in the visual occlusion experiments [34] is required to fairly quantify the proportion of simultaneous response transitions to familiar and novel images at odds with PE circuits.

A real possibility is that plasticity directions problematic for PE circuits could occur in PyC types beyond the canonical PPE and NPE neurons. The experiments which discovered Rad and Adams2 populations enriched for PPE and NPE responses [20] did not rule out the possibility of other PyC types in L2/3. It is possible that PyCs receiving both cRF and ecRF inputs obeying opposing plasticity rules are simply a third type of PyC in L2/3 of the cortex, accounting for a fraction of response transitions observed during visual occlusion [34]. Here, we argue for a microarchitecture of L2/3 that can give rise to effective connectivity mirroring PPE and NPE motifs through experience-dependent plasticity, but does not constrain the space of useful connectivity options before learning even starts. Ultimately, discriminating these accounts requires experimentally measuring apical and basal activation of many individual L2/3 neurons during visual occlusion. Prior work employing such approach on simple pattern-violating stimuli [33] found evidence compatible with our proposed model, showing that the difference between apical and somatic responses to pattern-violating stimuli grows with experience, reflecting opposing plasticity governing apical and somatic compartments. These and our findings run directly counter to dendritic formulations of predictive coding, where prediction errors are computed within single PyCs by subtracting the apical dendritic potential from somatic activity, and minimized by aligning apical and somatic responses with experience [65, 66].

The enhanced responses to novel scenes also admit a predictive coding interpretation. The effects we explain with generalized apical potentiation, could reflect an experience-dependent increase in confidence in the learned expectations, which would be expected to enhance otherwise unchanged positive prediction errors for novel stimuli [67]. Increases in attention or motivation driven by the training reward, would predict a similar enhancement of novel responses. Two observations from the experimental data narrow down this alternative. Firstly, familiarization did not increase responses to occluded noise images (Fig. 1I), ruling out nonvisual up-regulation through arousal, motivation or attention. Still, from a predictive coding perspective, a learned precision signal specific to natural scene statistics predicts exactly this dissociation [59]. Varying reliability of the stimulus statistics independently of familiarity would separate learned precision effects from generalized apical potentiation directly. The observed effects also do not require reward: while +O transitions are clearest in the rewarded Expert (Task) condition, they are still robustly present without reward (Fig. 2), suggesting reward amplifies rather than explains this form of plasticity, perhaps through diffuse neuromodulation.

Furthermore, our comparisons in Fig. 6 used the simplest possible implementation of PE circuits. A complete hPC implementation would innervate each PyC with connections from several

interneuron types providing separate subtractive and divisive inhibition, include L5 representation neurons and a hierarchy of areas, and let precision and confidence signals themselves be learned [59, 64, 67, 68]. Such circuits are significantly more expressive than the presented minimal version of CC, and many transitions not captured by the minimal L2/3 PPE and NPE motifs may be recoverable in them. Here, we kept the comparison at one PyC and one PV to retain a similar level of complexity at which our CC circuit operates (Fig. 5). Extending our CC model with L5 connectivity - allowing fairer comparison with established predictive coding circuits where L5 representation neurons are essential [59, 64, 67, 68] - is a natural next step for future work.

### 3.2 New hypotheses for experimental validation

Our simulated silencing experiments contribute concrete new predictions about mechanisms of L2/3 plasticity under visual occlusion, that can be tested with the same silencing experiments in-vivo. First, optogenetic silencing of local PV interneurons in L2/3 of expert mice V1 should selectively rescue the sensory responses of contextual responders. The extent to which it also rescues sensory responses of the -NO class will inform about the relative contribution of feedforward adaptation versus feedforward and lateral inhibition in expectation suppression of familiar sensory responses. Second, selective silencing of feedback signals from higher visual areas, should abolish contextual responses of expert PyCs and reduce responses of the +NO class below naive level. The residual sensory drive of +NO responders under FB silencing would be informative about the extent of long-term cRF plasticity. Furthermore, our hypothesis that a shorter-timescale feedforward adaptation process gates long-term feedforward plasticity predicts that sensory response trajectories should decay faster in broadly-tuned than in narrowly-tuned cells, and this decay rate should a) be measurable from the first few sessions of a familiarization protocol, reflecting the neuron’s cRF tuning, and b) should predict the sign of each cell’s long-term sensory response transition. This prediction requires no additional experimental perturbation, simply requiring tracking of sensory responses across the familiarization protocol, and would hold equally in a variant of our model where short-term adaptation gates learning of the lateral inhibitory weights while cRF tuning remains fixed. Finally, the most direct test of our account of predictive learning involving opposing compartment-specific plasticity rules would involve tracking of apical dendrites and somata of the same L2/3 PyCs across all stages of the visual occlusion paradigm, as has been previously done for pattern-violating stimuli [33]. The same experiments could also test whether the apical potentiation we propose generalizes across contexts, by comparing dendritic responses to familiar and novel scenes in the same neurons.

### 3.3 Limitations

While our work provides a detailed account of L2/3 predictive learning consistent with results of in-vivo visual occlusion experiments [34], several aspects of our model and methodology require attention and future refinement. A large share of the work presented here went into tuning initial conditions that could recover population-level distribution of transitions observed in the experiments (Fig. 4) - from finding concrete transition templates that

could tile the response change distribution space well, to adjusting proportions of neurons sampled from the different templates. In hindsight, this process involved significant manual input and introduced free parameters which could allow other initial configurations to reproduce the same results. Nevertheless this aspect remains clearly separable from the impactful contributions of our work. While the exact distribution of response changes in L2/3 PyCs is largely a function of population-level noise on the distribution of naive cRF, ecRF, and interneuron tuning, the core computational principles enabling the different transitions, arise from the plasticity rules and connectivity introduced by our model. Isolating the most robust, average response transitions, highlighting the expressive predictive learning plasticity enabled by our model calls for a mean-field treatment. One option could be to separately model one population for each zone of the response space, including silent, sensory-responsive, context-responsive, and jointly sensory-and-context-responsive populations, and modeling the most common transition for the "average" neuron from each population. Characterization of neural mechanisms responsible for the naive tuning distribution falls outside the scope of this work, and may need to consider different plasticity rules for development of cRFs and ecRFs during critical developmental periods [69].

A natural consequence of our bottom-up, data-driven modeling approach is that our unsupervised learning rules were derived to match data rather than optimize an objective function. This aspect of our model prevents any immediate functional applications of it to cognitive or machine learning tasks, which normative models of predictive learning in the brain excel at [2, 9, 59, 66, 68, 70, 71]. Still, it is possible that opposing unsupervised plasticity rules in L2/3 PyCs optimize an objective that could be recoverable analytically. Oja's Hebbian rule is a classic example of a purely local unsupervised learning rule that can be shown to perform a well-defined computation, namely extracting the first principal components of its input [72]. More radically, even without L2/3 optimizing any interpretable objective locally, the novelty and omission responses produced by our learning rules could serve as informative signals that other neural populations could use for credit assignment and predictive computation.

Furthermore, for simplicity, we ignored the role of feedback inputs to VIP interneurons providing PyC disinhibition. Together with direct PyC amplification, disinhibition is the dominant feedback motif in superficial layers of V1 [23], and thus should be included in any canonical model of L2/3 predictive learning. Including VIP interneurons in our circuit might also resolve transitions from complete silence into a robust +NO response that our CC model struggles to explain. These transitions carry more the signatures of disinhibition than of simple gain enhancement, and the anatomically supported [23] inclusion of VIP - and the related SST - interneurons would be the natural biological implementation of this mechanism.

### 3.4 Future directions

The structure of our model directly motivates several independent directions for future research. First, our CC plasticity rules use the rectified firing rate as the postsynaptic signal, which discards information about how far below firing threshold a cell sits. This

CC design choice forgoes the homeostatic mechanism in PE circuits which use signed voltage as the postsynaptic signal, allowing one plasticity rule to dynamically switch between Hebbian and anti-Hebbian modes to minimize prediction error (Fig. S2). Controlled characterization of the CC plasticity rules in both the rectified, and signed postsynaptic signal settings would yield insights on different regimes and dynamics of predictive learning in L2/3 circuits.

In the current CC model, short-term feedforward adaptation currently gates long-term feedforward depression. Alternatively, feedforward adaptation variables could be used to dynamically gate potentiation of somatic inhibitory synapses. Such variant of CC may not need plastic feedforward synapses at all, while preserving experience-dependent sharpening of sensory responses by selective suppression of broadly tuned cells. From a normative stance, modulating a cRF response by adjusting strength of surround suppression and excitatory feedback might be preferable to overwriting it. cRFs are built over critical developmental periods, potentially following development-specific plasticity rules [69], and define a hierarchy of feature detectors that higher areas rely on for complex tasks - so continuously making changes to cRFs based on experience might be costly and undo critical plasticity from development.

The final proposed extension is conceptual. In our simplified setting, the contextual feedback input is still image-specific, and only the plasticity generalizes. In reality, the arrangement should probably be the other way around, with general feedback signals spreading over several apical synapses, and plasticity being synapse specific. We expect such version of the model to result in functionally similar plasticity effects, while observing a simpler, more principled apical learning rule. Additionally, evidence that early visual cortex is modulated by high-level rather than low-level predictability [24, 27–29], together with the argument against learning from detailed generative prediction in neural networks [2, 11, 25, 26], favors feedback signals that are general to begin with, combined with local plasticity. Such forms of feedback would also enrich the comparison with PE circuits, since PE circuits might then show generalized plasticity of their responses to novel images which share contextual inputs with familiar images.

## 4 METHODS

### 4.1 Visual occlusion mouse experiments

All experimental methods, results and statistics for the two-photon imaging experiments during the visual occlusion paradigm are described in detail in previous work [34]. Here, we do not introduce new experiments or data, but simply collect and present previously published results (Figs. 1 & S1), or perform new analyses and plots of published experimental data pertinent to design of the circuit model (Figs. 2, 4, 6, S3), providing relevant details in Figure captions.

### 4.2 Circuit model

We modeled the opposing plasticity in L2/3 PyCs using a simplified rate-based circuit model outlined in Figure 3A, consisting of a single two-compartment L2/3 PyC  $y$  and a single-compartment L2/3 PV interneuron  $p$  providing feedforward divisive inhibition.

Overall, the firing rate of L2/3 PyC  $y$  was modeled as a nonlinear leaky integrator of the apically-modulated somatic activation  $\gamma b$ ,

apical drive  $\beta$ , and a small random baseline current  $I_y \sim N(0, \sigma_y^2)$ .

$$\tau_y \frac{dy(t)}{dt} = -y(t) + \phi \left( \frac{\gamma(t)b(t) + \beta(t)}{1 + \kappa \cdot w_{LAT}(t)p(t)} + I_y(t) - a_f(t) \right) \quad (4)$$

The basal / somatic compartment  $b(t) = \mathbf{w}_{FF}(t)^T \mathbf{x}(t)$  of the PyC received feedforward inputs  $\mathbf{x}(t) \in \{0, 1\}^3$  via dedicated basal feedforward synapses  $\mathbf{w}_{FF}(t) \in [0, 1]^3$  for feedforward inputs corresponding to each of the experimental stimuli. The integrated feedforward and contextual inputs were normalized by divisive feedforward inhibition [58] from the PV cell  $p$  via an inhibitory somatic synapse  $w_{LAT}(t) \in [0, 1]$ , with parameter  $\kappa = 10$  setting the maximum possible scale of divisive inhibition. The PyC also accumulated adaptation currents  $a$  with persistent stimulation, leading to biologically realistic short-term (within-trial) adaptation of neural responses as result of the fast adaptation current  $a_f$ , and leading to altered long-term feedforward plasticity as result of the slow adaptation current  $a_s$ .

$$\tau_a \frac{da(t)}{dt} = -a(t) + y(t); \quad a \in (a_f, a_s); \quad \tau_{a_s} \ll \tau_{a_f} \ll \tau_y \quad (5)$$

The PyC firing was gated by a rectified linear activation function,  $\phi(u|\theta_y) = \mathbb{I}[u > \theta_y] \cdot u$ , with firing threshold  $\theta_y = 0.08$  for all cells.

The apical dendritic compartment  $d$  of the PyC received contextual inputs  $\mathbf{c} \in \{0, 1\}^3$ , with dedicated apical feedback synapses  $\mathbf{w}_{FB} \in [0, 1]^3$  for contextual inputs corresponding to the experimental stimuli,  $d(t) = \mathbf{w}_{FB}(t)^T \mathbf{c}(t)$ . To account for nonlinear nature of apical feedback which can amplify somatic activity [44], but also drive somatic firing when feedback is sufficiently strong [45], we treated the apical multiplicative gain  $\gamma$  and apical drive  $\beta$  as separate terms.

$$\gamma(t) = \sigma(d(t)); \quad \beta(t) = \psi(d(t)) \quad (6)$$

Apical gain  $\gamma$  was modeled with a thresholded, shifted sigmoid function  $\sigma(u|g, k) = \max\{1, 1 + \frac{g}{1 + \exp(-k(u - \theta_g))} - \frac{g}{2}\}$ , to ensure that apical feedback could enhance, but not suppress somatic activity, with parameters  $g$  and  $k$  controlling the apical gain ceiling, and steepness of the gain function. Apical feedback acted as multiplicative gain on somatic activity as soon as apical activation crossed a threshold  $\theta_g = 0.07$  for all cells. Apical drive  $\beta$  was modeled with a shifted rectified linear function  $\psi(u|\theta_d) = \max\{0, u - \theta_d\}$ , only contributing to neuronal activity once apical activation crossed a threshold  $\theta_d$ . The apical activation necessary to for gain modulation of somatic activity was substantially lower than the apical activation necessary for apical drive  $\theta_g < \theta_d$ .

$$\tau_p \frac{dp(t)}{dt} = -p(t) + \phi \left( \mathbf{w}_{PV}^T \mathbf{x}(t) + w_{pLAT} y(t) + I_p(t) \right). \quad (7)$$

We modeled feedforward divisive inhibition, with a point PV interneuron  $p$  receiving the same feedforward input  $\mathbf{x}$  as the pyramidal cell, with distinct synapses for each of the feedforward inputs  $\mathbf{w}_{PV} \in [0, 1]^3$ . The PV cell also received a lateral connection from the PyC  $w_{pLAT} \in [0, 1]$ , as well as a small random baseline current  $I_p \sim N(0, \sigma_p^2)$ . The firing rate of the PV  $p$  was also modeled as a nonlinear leaky integrator, with the same nonlinearity as PyC. The firing rates of both PyC and PV were clipped to 1.0, representing the maximum possible firing rate for each cell type.

**4.2.1 Plasticity rules** In our model, PyC synapses undergo plasticity according to three different learning rules highlighted by the different colors in Figure 7A, while PV synapses do not change with experience. Updates for all potentiating weights are softly bounded by a factor of  $(1 - w)$ , and by a factor of  $w$  for adapting weights, such that synaptic weights range between 0-1.

$$\frac{d\mathbf{w}_{FF}(t)}{dt} = -\tilde{\eta}_{FF}(t) \cdot y(t) \cdot \mathbf{x}(t) \odot \mathbf{w}_{FF}(t); \quad \tilde{\eta}_{FF}(t) = \zeta \cdot \eta_{FF} \cdot a_s^2(t) \quad (8)$$

The PyC feedforward synapses  $\mathbf{w}_{FF}$  undergo selective soft-bounded depression as function of postsynaptic activity  $y$ , at a dynamic rate  $\tilde{\eta}_{FF}$ . Based on the increased selectivity of NO responses to both familiar and novel images after familiarization (Fig. 1D,J), we hypothesized that the effective learning rate  $\tilde{\eta}_{FF}$  scales with breadth of naive feedforward tuning of L2/3 PyCs. In our model broadly tuned PyC's (higher weights to 2/3 of the cRF inputs) adapt more rapidly than narrowly tuned PyC's (higher weight only to 1/3 of the cRF inputs). This phenomenon emerges due to the slow timescale  $\tau_{a_s}$  which makes  $a_s$  accumulate over several independent stimulus presentations, and thus, through dynamic modulation of  $\eta_{FF}$ , on average result in  $\tilde{\eta}_{FF}$  being consistently higher in broadly compared to narrowly tuned cells. We experimented with several ways of modulating  $\eta_{FF}$  with  $a_s$ , and found that quadratic dependence on  $a_s$ , paired with a shared, fixed normalization factor  $\zeta$  produced dynamic learning rates that clearly distinguished plasticity in broadly and narrowly-tuned cells.

$$\frac{d\mathbf{w}_{FB}(t)}{dt} = \eta_{FB} \frac{\lambda}{y(t) + \lambda} (\mathbf{S}^c \mathbf{c}(t)) \odot (1^3 - \mathbf{w}_{FB}(t)) \quad (9)$$

Conversely, PyC apical feedback synapses  $\mathbf{w}_{FB}$  broadly strengthen in an inverse relationship with postsynaptic activity  $y$ , at a rate  $\eta_{FB}$ . The  $\mathbf{S}^c \in \mathbb{R}^{3 \times 3}$  matrix determines the selectivity of feedback plasticity—its diagonal elements determine the strength of context-selective strengthening, and off-diagonal elements specify strength of general cross-context strengthening. In our simulations we set  $\lambda = 1$ .

$$\frac{dw_{LAT}(t)}{dt} = \eta_{LAT}(y(t) \cdot p(t))(1.0 - w_{LAT}(t)) \quad (10)$$

Finally, the weight of the inhibitory synapse  $w_{LAT}$  onto the PyC evolves according to a soft-bounded Hebbian plasticity rule, with learning rate  $\eta_{LAT}$ .

**4.2.2 Simulation details** We studied PyC responses to two familiar inputs ( $\mathbf{E}_1, \mathbf{E}_2$ ) and one novel input ( $\mathbf{E}_3$ ). Each input matrix representing non-occluded (NO) inputs  $\mathbf{E}_i^{NO} \in \{0, 1\}^{3 \times 2}$  consisted of 2 stacked one-hot vectors,  $\mathbf{x}_i^{NO}, \mathbf{c}_i \in \{0, 1\}^3$  representing image-specific feedforward and contextual inputs respectively. Occluded images  $\mathbf{E}_i^O$  were modeled by replacing the image-specific feedforward input with a vector of zeros,  $\mathbf{x}_i^O = \mathbf{0}^3$ . We simulated our model in discrete time, yielding  $T$  frames per each trial, consisting of  $\frac{1}{4}T$  pre-stimulus ITI frames,  $\frac{1}{4}T$  stimulus presentation frames, and  $\frac{1}{2}T$  post-ITI frames. During the ITI both the model input consisted of only zeros  $\mathbf{E}^{ITI} = \mathbf{0}^{3 \times 2}$ , representing absence of visual input.

We followed the same experimental protocol as the real experiment (Figure 3B), first testing responses of naive neurons to both NO and O versions of all inputs ( $\mathbf{E}_1^{NO/O}, \mathbf{E}_2^{NO/O}, \mathbf{E}_3^{NO/O}$ ). Subsequently, we familiarized neurons with the NO versions of the familiar images



only ( $E_1^{NO}, E_2^{NO}$ ). After familiarization, we tested the responses of expert neurons to all inputs again ( $E_1^{NO/O}, E_2^{NO/O}, E_3^{NO/O}$ ). Plasticity rules were only activated during the familiarization phase—in both naive and expert testing sessions, plasticity was turned off to isolate the effect of plasticity during the familiarization phase. In the expert testing session we included three additional perturbation conditions during which we tested the neuronal response to all inputs again—silencing the PVs (*PV silencing*); silencing the contextual feedback inputs (*FB silencing*); and silencing both PVs and contextual feedback (*PV & FB silencing*). These perturbations allowed us to better understand the role of different components of the model, and generate hypotheses for future experiments which could perform these perturbations using in-vivo optogenetics.

We modeled the variety of neuronal transitions observed in-vivo by constructing a set of average transitions (transition templates), that captured important phenomenological transitions observed in the data and enabled by the model. We defined a set of PyC and PV parameters shared across transition templates (Table 1), as well as, a second set of PyC and PV parameters defining each transition template (Table 2).

To fairly represent a population of neurons with diverse image preferences, we made feedforward tuning width and strength a defining characteristic of each transition template (broad cells were tuned to 2/3 inputs, and narrow cells to 1/3). PyCs could have broad or narrow feedforward tuning  $w_{FF}$ . PVs were always modeled with broad feedforward tuning  $w_{PV}$  with less extreme differences between synapse strength for preferred and non-preferred inputs, to reflect their role in local surround suppression and divisive normalization. Each transition template defined the average tuning strength of preferred  $\tilde{w}^{pref}$  and non-preferred  $\tilde{w}^{non-pref}$  synapses for PyC and PV separately. Crucially, the combination of which synapses were preferred vs non-preferred was sampled with uniform probability for each PyC and PV independently. Because we only used three inputs, broadly-tuned PyCs were guaranteed to share at least one preferred input with the (broadly-tuned) PVs. Narrow-tuned PyCs had a 66.67% chance to share their preferred input with the PV. We chose this approach to guarantee that all equally-likely PyC-PV stimulus preference combinations are proportionately represented in the sampled sub-population defined by each transition template. After determining the average tuning strength and sampling the tuning preference of each neuron  $i$ , we obtained the its average weight vector  $\tilde{w}_u^i = \{\tilde{w}_u^{pref_i}\} \cup \{\tilde{w}_u^{non-pref_i}\}$ ,  $u \in (FF, PV)$ . Average feedback strength  $\tilde{w}_{FB}$  in naive state was modeled as diffuse, equally affecting all three contextual synapses according to feedback strength specified by the transition template.

To reflect variability in basal and apical synaptic strength in L2/3 PyC neurons across tuning configurations, for each neuron  $i$  (whose average weight vectors we obtained according to the procedure above), we independently added Gaussian noise to elements of  $\tilde{w}_{FF}, \tilde{w}_{FB}$ , with variance  $\sigma_u^2$  proportional to the average weight according to proportionality parameters  $r_u, u \in (FF, FB)$  shared across all neurons. A minimum amount of variance  $\sigma_{u_{min}}^2$  was enforced to maintain variability around weak weights,  $\sigma_u^2 = \max\{r_u|\tilde{w}_u|, \sigma_{u_{min}}^2\}$ . The actual sampled weights  $w_u = \tilde{w}_u + N(0, \sigma_u^2)$ ,  $u \in (FF, FB)$  used in the simulations for each sampled neuron

were finally clipped to be non-negative and below a maximum threshold  $w_u^{MAX}$  to prevent extreme outliers. Since noise was only injected into feedforward and feedback weights of the PyC, the other template-defined weights exactly parametrized the PV-related weights for all neurons drawn from that template  $w_u = \tilde{w}_u, \forall u \in (PV, LAT, pvLAT)$ .

To reflect variability in dendritic morphology of L2/3 PyC neurons, for each neuron we multiplied the apical gain ceiling  $\tilde{g}$  specified by each transition template with lognormal noise  $g = \tilde{g} \cdot \exp(N(0, r_g))$ , and added Gaussian noise to apical drive threshold  $\tilde{\theta}_d$  specified by each transition,  $\theta_d = \tilde{\theta}_d + N(0, \max\{r_{\theta_d} \cdot |\tilde{\theta}_d|, \sigma_{\theta_{d_{min}}}^2\})$ . Similar to the weights, both parameters were clipped to a broad range to prevent outlier artifacts.

After constructing the transition templates and adjusting proportions of samples drawn from each transition template to match the naive familiar and naive novel scatter-plots, we could sample from this population of neurons and study the NO/O response transitions that occurred on population level across random samples as a result of familiarization.

Each neuron’s firing rate was standardized based on the mean and standard deviation of the neuron’s firing rate across all the pre-stimulus periods in the naive testing phase.

We ran our simulations in discrete time, using Euler integration, yielding a discrete-time leaky integration version of our differential equations for the PyC firing rate  $y$ , PV firing rate  $p$ , and adaptation current  $a$ .

$$u_{t+1} = (1 - \alpha_u)u_t + \alpha_u \tilde{u}_t; \quad u \in (y, p, a); \quad \tau_u = \frac{\Delta t}{\alpha_u}, \quad (11)$$

where  $\tilde{u}_t$  is the instantaneous activation. Here the leak-rate is determined by the constants  $\alpha_u$ , which are related to the continuous time-constants  $\tau_u$  via the time-discretization  $\Delta t$ .

### 4.3 Comparison with matched predictive-coding circuits

We compared the context-contrasting (CC) circuit with minimal positive and negative prediction-error circuits (PPE and NPE). The two predictive-coding models used the same parameter rows and differed only by exchanging sensory input  $x(t)$  and contextual input  $c(t)$ . In the PPE circuit, the PyC received direct sensory excitation  $x(t)$  and the PV cell received contextual excitation  $c(t)$ .

$$e_+(t) = w_E^T x(t) - w_{LAT} p_+(t) \quad (12)$$

$$\tau_p \frac{dp_+(t)}{dt} = -p_+(t) + \phi(w_{PV}^T c(t)) \quad (13)$$

In the NPE circuit, the PyC received direct contextual excitation  $c(t)$  and the PV cell received sensory excitation  $x(t)$ .

$$e_-(t) = w_E^T c(t) - w_{LAT} p_-(t) \quad (14)$$

$$\tau_p \frac{dp_-(t)}{dt} = -p_-(t) + \phi(w_{PV}^T x(t)) \quad (15)$$

For both circuits, the rectified PyC response was

$$\tau_y \frac{dy_{\pm}(t)}{dt} = -y_{\pm}(t) + \phi(e_{\pm}(t) + I_y - a_f(t)) \quad (16)$$

where  $a_f$  was the short term adaptation current, and  $I_y$  the baseline noise also used in the context contrasting circuit.

Only the direct excitatory PyC synapses  $w_E$  were plastic: representing basal feedforward synapses  $w_{FF}$  in PPEs and apical feedback synapses  $w_{FB}$  in NPEs. Both circuits used the same soft-bounded anti-Hebbian rule,

$$\delta(t) = -\eta e_{\pm}(t)u(t), \quad \Delta w_{E,i}(t) = \begin{cases} \delta_i(t)(1 - w_{E,i}(t)), & \delta_i(t) \geq 0, \\ \delta_i(t)w_{E,i}(t), & \delta_i(t) < 0, \end{cases} \quad (17)$$

with  $u = x$  for PPEs and  $u = c$  for NPEs. The inhibitory weight  $w_{LAT}$  subtracting PV activity from the PyC, and the excitatory weights driving PV activity  $w_{PV}$  were fixed. A single learning rate was chosen as the smallest value achieving  $e_{\pm}$  convergence (with  $\pm 0.005$  tolerance) within the full-protocol (same number and duration of trials as context contrasting simulation).

Because of the greater simplicity and symmetry between PPE and NPE circuits, construction of the PPE and NPE populations parameters used one shared parameter pool. PyC excitatory strength, PV excitatory strength and  $w_{LAT}$  were sampled independently over the ranges in Table 3; PyC and PV preferred-channel sets were also sampled independently (just like for the CC circuits). The same set of parameters could then mapped to either PPE or NPE by swapping the sensory/context inputs. Thus, unlike the context-contrasting population, the predictive-coding population was not fitted to the empirical scatter plots or proportions. Simulation, and evaluation followed the same procedure as CC.

## 5 RESEARCH IN CONTEXT STATEMENT

### 5.1 Evidence before this study

Recent years have seen convergence between neuroscience and machine learning regarding the idea that predictive learning and computation form the bedrock of intelligent systems. The prevalent theory about the neural instantiation of these computational principles is hierarchical predictive coding - which asserts the brain computes errors between its predictions and sensory experience to guide the process of inferring and learning the latent causes of its observations. While neuroscientists have found abundant experimental evidence supporting hypotheses stemming from predictive coding, many assumptions about its low-level implementation remain contested. Notably, the reliance on prediction-driven subtractive interneuron activity in L2/3 circuits, and the assumption about clean division of labor between specialized types of error signaling cells in L2/3, have attracted criticism.

### 5.2 Added value of this study

In this work, we propose an account of unsupervised predictive learning in L2/3 pyramidal cells, which is consistent with anatomical evidence and does not rely on prediction-driven inhibition or specialized error cells. We find that our model reproduces the hallmarks of predictive learning observed in mice exposed to the visual occlusion paradigm, and highlights experimentally-observed modes of plasticity problematic for L2/3 circuits obeying assumptions of classical hierarchical predictive coding.

### 5.3 Implications of all the available evidence

This work contributes to the growing line of evidence for departure from the idealized hierarchical predictive coding framework. Nevertheless, many experimental and theoretical observations from this and prior work remain consistent with the broader predictive coding literature. Gathering detailed experimental evidence targeted at concrete, biological and computational assumptions of established and emerging theories will be critical in formulating of a new theory of predictive learning in the brain.

## 6 SUMMARY FOR THE GENERAL PUBLIC

Anticipating what comes next, and making sense of things we can only partly see, are two indispensable abilities each human and animal relies on. Walking home on a foggy night we might make out only a few steps ahead, yet we still know where the pavement continues, where the gate is, and what to expect around each corner. Our brain works not only with what our eyes can see but also uses context, other senses and past experience to form rich percepts of the environment that guide our actions.

Exactly how the brain does this is still debated. The leading theory holds that the brain contains special "error" cells whose job is to compare what was expected to what was actually seen, and report only the surprising difference. Two kinds of surprise can be reported by such cells: something novel and unexpected can happen, like driving on an empty road and suddenly seeing a deer in the headlights; or something expected can fail to happen, like coming home to find your driveway empty and your car stolen. Despite the elegance of this theory, the wiring of the visual brain does not seem fit for calculating differences: the feedback connections that should carry expectations mostly enhance activity of cells reporting what was seen rather than subtract from it.

In our work, we took a different route in trying to understand how the brain might learn to communicate about novelty and omission without calculating differences. Starting from recordings of individual cells in the visual brain of mice, made while the animals viewed familiar and new photographs with parts of the photographs missing, we built a simple computer model of a brain cell. In this model, connections arriving at different parts of the same cell change in opposite directions as a scene becomes more familiar: the connections carrying information about what was seen weaken, while the connections carrying information about what was expected strengthen.

That imbalance is enough. Simulations of cells from our model reproduce both the surge of activity when something genuinely new and unexpected appears, and the activity that arises when an expected part of a scene goes missing - without relying on specialised "error" cells.

This matters beyond abstract arguments about whether error-reporting cells exist. When the brain's sense of surprise breaks down, ordinary things can start to feel unexpectedly novel, and extremely unusual things can start to feel anticipated and missing - a process thought to contribute to hallucinations and delusions. By improving our understanding of how brains might develop anticipatory and surprise signals, we move closer towards helping people with brains in which those signals start to misbehave.

| Symbol                                                                                           | Value                                                           | Meaning                                                                     |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------|
| <i>Neural dynamics</i>                                                                           |                                                                 |                                                                             |
| $\alpha_y$                                                                                       | 0.05                                                            | PyC leak rate ( $\tau_y = \Delta t / \alpha_y$ )                            |
| $\alpha_p$                                                                                       | 0.5                                                             | PV leak rate ( $\tau_p = \Delta t / \alpha_p$ )                             |
| $\theta_y$                                                                                       | 0.08                                                            | Soma firing threshold (PyC and PV)                                          |
| $\kappa$                                                                                         | 10.0                                                            | Divisive normalization scale                                                |
| $\sigma_p$                                                                                       | 0.075                                                           | PV baseline current s.d.                                                    |
| $\theta_g$                                                                                       | 0.05                                                            | Apical gain onset threshold                                                 |
| $\alpha_{af}$                                                                                    | 0.01                                                            | Fast adaptation rate ( $\tau_{af} = \Delta t / \alpha_{af}$ )               |
| <i>Simulation protocol</i>                                                                       |                                                                 |                                                                             |
| $T$                                                                                              | 400                                                             | Discrete time steps per trial (ITI + stim + post-ITI)                       |
| $N_{train}$                                                                                      | 7                                                               | Familiarization (training) trials per familiar image                        |
| $N_{test}$                                                                                       | 5                                                               | Testing trials per image (naive and expert phases)                          |
| <i>Plasticity (rates listed for 200-step reference; scaled by <math>200/T</math> at runtime)</i> |                                                                 |                                                                             |
| $\eta_{FF}$                                                                                      | 0.0155                                                          | FF depression rate                                                          |
| $\eta_{FB}$                                                                                      | 0.00065                                                         | Apical feedback potentiation rate                                           |
| $\eta_{LAT}$                                                                                     | 0.0300                                                          | Lateral inhibitory Hebbian rate                                             |
| $\alpha_{as}$                                                                                    | 0.0025                                                          | Slow adaptation rate ( $\tau_{as} = \Delta t / \alpha_{as}$ )               |
| $\zeta$                                                                                          | 2000                                                            | Magnitude normalization constant for $a_s$ affecting dynamic $\tilde{\eta}$ |
| $\lambda$                                                                                        | 1                                                               | FB anti-Hebbian denominator constant                                        |
| $S^c$                                                                                            | $0.6 \cdot \mathbf{I} + 0.2 \cdot (\mathbf{11}^T - \mathbf{I})$ | FB plasticity specificity.                                                  |
| <i>Population variability in basal / apical weights</i>                                          |                                                                 |                                                                             |
| $r_{FF}$                                                                                         | 0.28                                                            | FF relative-noise proportionality                                           |
| $\sigma_{FF}^{min}$                                                                              | 0.016                                                           | FF variance floor                                                           |
| $w_{FF}^{MAX}$                                                                                   | 0.35                                                            | FF clip max                                                                 |
| $r_{FB}$                                                                                         | 0.40                                                            | FB relative-noise proportionality                                           |
| $\sigma_{FB}^{min}$                                                                              | 0.016                                                           | FB variance floor                                                           |
| $w_{FB}^{MAX}$                                                                                   | 0.65                                                            | FB clip max                                                                 |
| <i>Population variability in dendritic parameters</i>                                            |                                                                 |                                                                             |
| $r_g$                                                                                            | 0.18                                                            | Log-normal s.d. on apical gain ceiling $g$                                  |
| $g_{min}$                                                                                        | 1.5                                                             | Post-noise lower clip on $g$ (upper clip $g^{MAX}$ is per-template)         |
| $r_{\theta_d}$                                                                                   | 0.12                                                            | Apical drive threshold $\theta_d$ noise proportionality                     |
| $\sigma_{\theta_d}^{min}$                                                                        | 0.05                                                            | Drive-threshold variance floor                                              |
| $\theta_d$ clip                                                                                  | [0.0, 1.5]                                                      | Post-noise clip on $\theta_d$                                               |

**Table 1: Parameters shared across all transition templates.**

| Template        | W | FB | $\pi$ | $\tilde{w}_{FF}^{p/n}$ | $\tilde{w}_{FF}^{n/n}$ | $\tilde{w}_{PV}^{p/n}$ | $\tilde{w}_{PV}^{n/n}$ | $\tilde{w}_{FB}$ | $\tilde{g}$ | $g^{MAX}$ | $k$ | $\tilde{\theta}_d$ | $\tilde{w}_{LAT}$ | $\tilde{w}_{pLAT}$ | $\sigma_y$ |
|-----------------|---|----|-------|------------------------|------------------------|------------------------|------------------------|------------------|-------------|-----------|-----|--------------------|-------------------|--------------------|------------|
| silentFF-noFB   | B | ×  | 0.012 | 0.010                  | 0.010                  | 0.240                  | 0.050                  | 0.004            | 3.8         | 6.4       | 5.0 | 0.16               | 0.040             | 0.08               | 0.18       |
| silentFF-midFB  | B | ✓  | 0.035 | 0.010                  | 0.010                  | 0.880                  | 0.280                  | 0.075            | 2.8         | 6.4       | 5.0 | 0.040              | 0.900             | 0.20               | 0.18       |
| silentFF-highFB | B | ✓  | 0.028 | 0.010                  | 0.010                  | 0.880                  | 0.280                  | 0.300            | 2.0         | 6.4       | 5.0 | 0.160              | 0.850             | 0.05               | 0.26       |
| weakFF-noFB     | B | ×  | 0.060 | 0.105                  | 0.006                  | 0.240                  | 0.050                  | 0.004            | 3.8         | 6.4       | 5.0 | 0.16               | 0.035             | 0.10               | 0.32       |
| midFF-noFB      | B | ×  | 0.180 | 0.145                  | 0.006                  | 0.660                  | 0.170                  | 0.004            | 3.8         | 6.4       | 5.0 | 0.16               | 0.050             | 0.10               | 0.32       |
| highFF-noFB     | B | ×  | 0.380 | 0.190                  | 0.008                  | 0.660                  | 0.170                  | 0.004            | 3.8         | 6.4       | 5.0 | 0.16               | 0.050             | 0.10               | 0.32       |
| midFF-midFB     | B | ✓  | 0.080 | 0.145                  | 0.006                  | 0.660                  | 0.170                  | 0.075            | 6.0         | 7.5       | 7.0 | 0.080              | 0.240             | 0.05               | 0.22       |
| strongFF-midFB  | B | ✓  | 0.110 | 0.190                  | 0.008                  | 0.660                  | 0.170                  | 0.075            | 6.4         | 8.0       | 7.0 | 0.080              | 0.240             | 0.05               | 0.22       |
| strongFF-midFB  | B | ✓  | 0.110 | 0.190                  | 0.008                  | 0.880                  | 0.280                  | 0.075            | 7.2         | 9.0       | 5.0 | 0.080              | 0.240             | 0.05               | 0.22       |
| strongFF-weakFB | B | ✓  | 0.280 | 0.190                  | 0.008                  | 0.310                  | 0.130                  | 0.050            | 7.8         | 9.0       | 8.0 | 0.240              | 0.360             | 0.05               | 0.22       |
| tinyFF-weakFB   | N | ✓  | 0.220 | 0.065                  | 0.004                  | 0.045                  | 0.010                  | 0.050            | 8.0         | 9.5       | 9.0 | 1.08               | 0.030             | 0.03               | 0.10       |
| weakFF-weakFB   | N | ✓  | 0.540 | 0.105                  | 0.006                  | 0.045                  | 0.010                  | 0.050            | 8.4         | 10.0      | 9.0 | 1.08               | 0.030             | 0.03               | 0.10       |
| midFF-noFB      | N | ×  | 0.010 | 0.145                  | 0.006                  | 0.240                  | 0.050                  | 0.004            | 7.5         | 8.0       | 9.0 | 1.08               | 0.030             | 0.03               | 0.10       |

**Table 2: Per-template parameters.** W: PyC feedforward tuning width (B = broad, 2/3 preferred FF channels; N = narrow, 1/3). FB: whether cell receives relevant contextual FB / not (✓/×).  $\pi$ : mixture weight (fraction of the sampled population; sums to 2.043 and is renormalized at sampling time).  $\tilde{w}_{FF}^{p/n}, \tilde{w}_{PV}^{p/n}$ : mean preferred / non-preferred basal weights for PyC and PV; the preferred-channel index set is drawn uniformly per cell and independently for PyC and PV.  $\tilde{w}_{FB}$ : mean feedback weight, applied uniformly to all context channels a cell receives for cells receiving relevant contextual FB (FB = ✓).  $\tilde{g}, g^{MAX}$ : apical gain ceiling center and its upper post-noise clip ( $g_{min} = 1.5$  for all templates).  $k$ : apical-gain sigmoid steepness.  $\tilde{\theta}_d$ : apical drive threshold center.  $\tilde{w}_{LAT}, \tilde{w}_{pLAT}$ : initial PV→PyC and PyC→PV lateral weights.  $\sigma_y$ : PyC baseline current s.d. Parameters not listed inherit the defaults from Table 1.

| Quantity                | PPE                        | NPE                        | Shared setting                                             |
|-------------------------|----------------------------|----------------------------|------------------------------------------------------------|
| Direct PyC input        | sensory <b>x</b>           | context <b>c</b>           | tuned weight $\mathcal{U}_{\text{strat}}$ (0.05, 0.95)     |
| PV input                | context <b>c</b>           | sensory <b>x</b>           | tuned weight $\mathcal{U}_{\text{strat}}$ (0.05, 0.95)     |
| Plastic synapse         | $\mathbf{w}_{FF}$          | $\mathbf{w}_{FB}$          | all other synapses fixed                                   |
| PyC/PV tuning width     | 1 or 3 of 3                | 1 or 3 of 3                | preferred channels sampled independently; untuned weight 0 |
| $w_{LAT}$               | PV→PyC                     | PV→PyC                     | $\mathcal{U}_{\text{strat}}$ (0.05, 0.95)                  |
| Rate updates            | $\alpha_y = 0.05$          | $\alpha_y = 0.05$          | $\alpha_p = 0.5, \alpha_a = 0.01$                          |
| Baseline current        | $\mathcal{N}(0.2, 0.30^2)$ | $\mathcal{N}(0.2, 0.30^2)$ | PV noise = 0                                               |
| Plasticity rate         | $\eta = 0.3$               | $\eta = 0.3$               | calibrated at 400 steps and $ e  \leq 0.005$               |
| Population and protocol | 300 cells                  | matched 300 cells          | seed 7151; 5 test and 7 training trials per familiar image |

**Table 3: Matched PPE/NPE model parameters. Stratified uniform sampling covered each interval evenly across the population. Every sampled row was instantiated once as PPE and once as NPE.**

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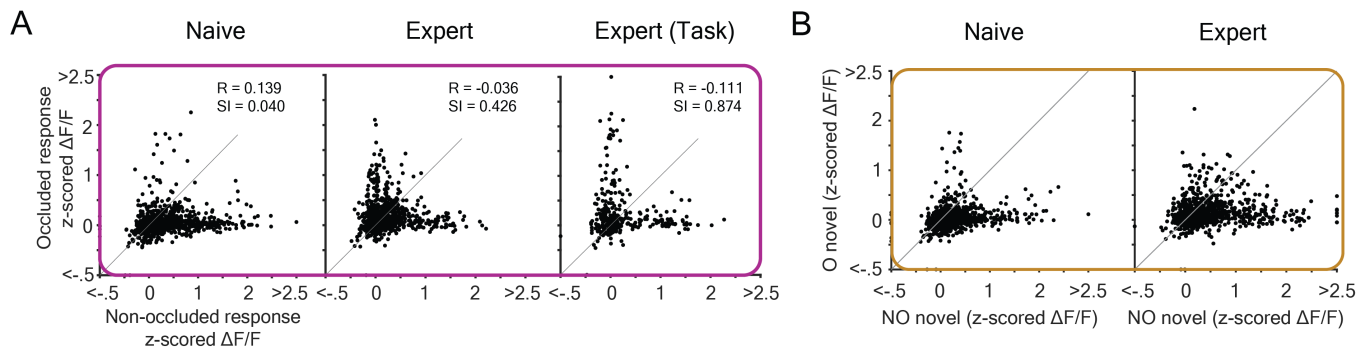
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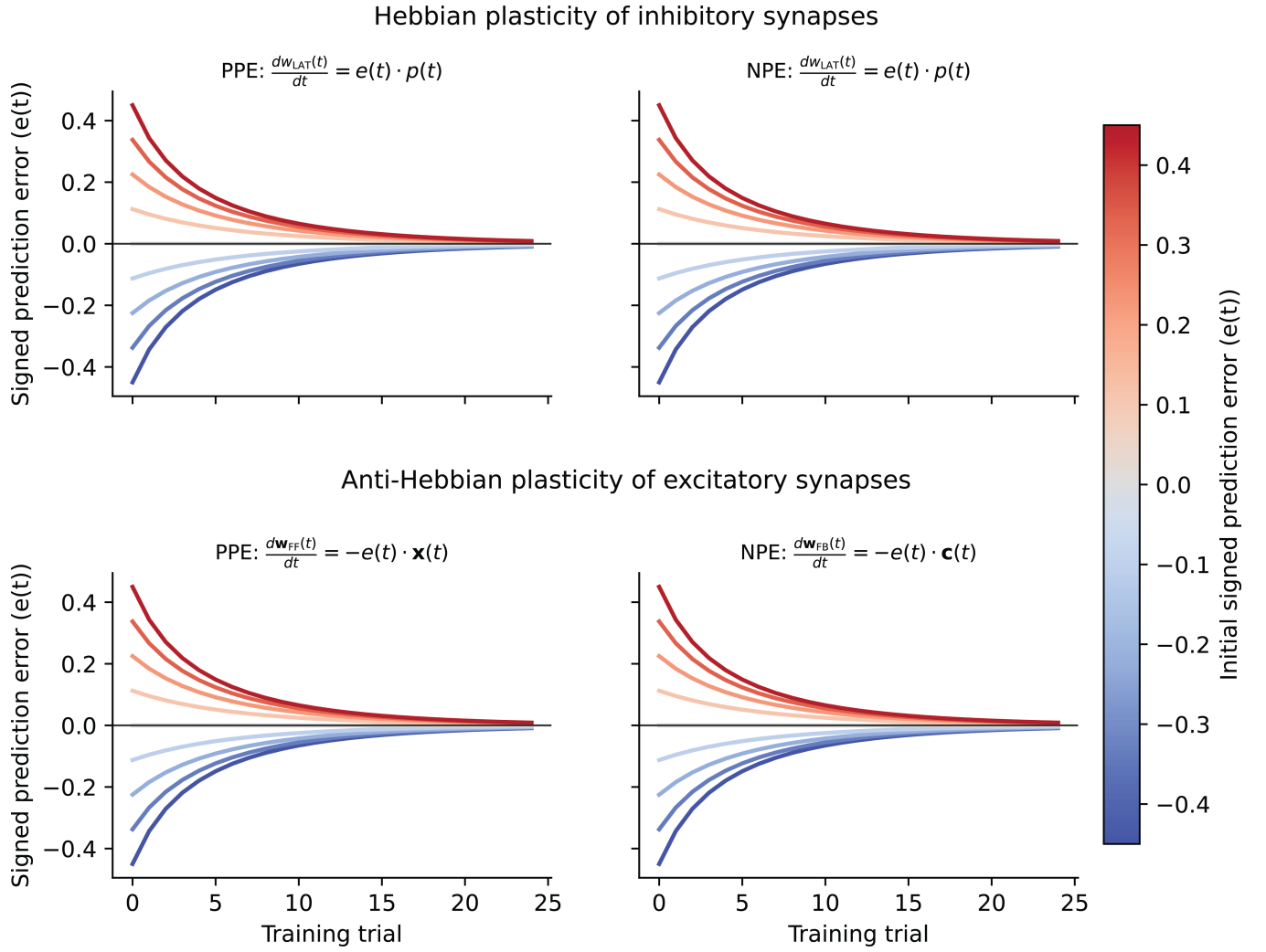


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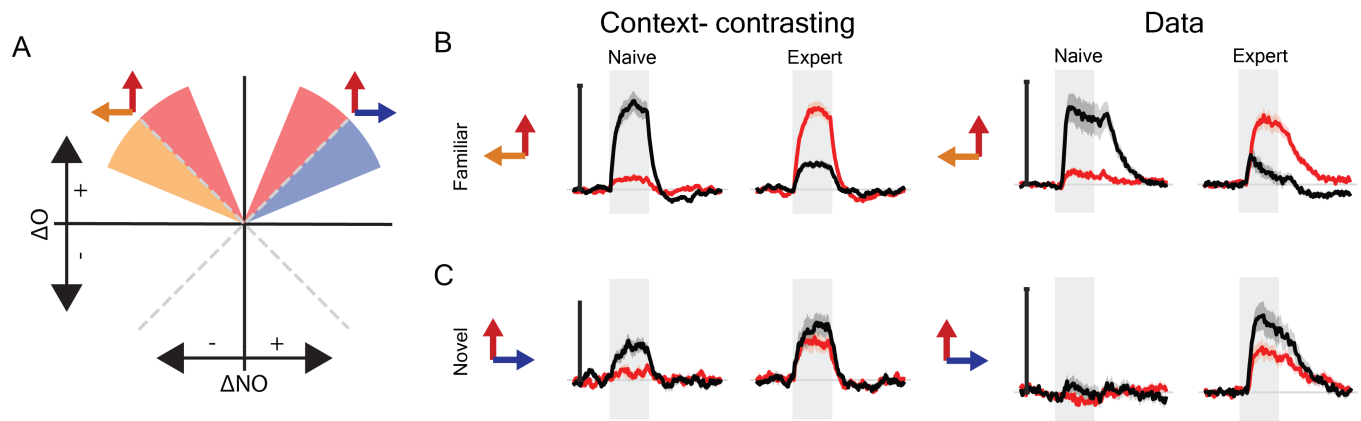
## 8 EXTENDED DATA



**Extended Data Figure 1: Familiarization increases separation of sensory and omission responses for expected stimuli, but decreases separation of sensory and omission responses for novel stimuli.** **A)** Scatter plot showing average z-scored  $\Delta F/F$  responses ( $t = 0.2-1$  s) to the non-occluded (x-axis) and occluded (y-axis) familiar natural images across conditions.  $N = 841$  neurons (Naive), 884 neurons (Expert) and 300 neurons (Expert (Task)) from 6 mice.  $R$ , correlation coefficient;  $SI$ , separation index. **B)** Scatter plot showing average z-scored  $\Delta F/F$  responses to non-occluded (x-axis) and occluded (y-axis) novel natural images in naive mice (left) and expert mice (right).



**Extended Data Figure 2: Hebbian plasticity of inhibitory PyC synapses and Anti-Hebbian plasticity of excitatory PyC synapses both minimize prediction error in PPE and NPE circuits.** (Top row) Simulation of PPE and NPE circuits with all weights frozen at 0.5, except  $w_{LAT}$  which is initialized at different initial strengths and evolves according to Hebbian plasticity between the post-synaptic membrane potential of the PyC, signifying the signed prediction error  $e(t)$ , and the presynaptic firing rate of the PV  $p(t)$ . (Bottom row) Simulation of PPE and NPE circuits with all weights frozen at 0.5, except for  $w_{FF}$  for PPE and  $w_{FB}$  for NPE, which are initialized at different initial strengths and evolve according to Anti-Hebbian plasticity between the post-synaptic membrane potential of the PyC  $e(t)$ , and the presynaptic excitatory input signal (feedforward input  $\mathbf{x}(t)$  for PPE, and contextual input  $\mathbf{c}(t)$  for NPE).



**Extended Data Figure 3: Context-contrasting reproduces simultaneous plasticity of cRF and ecRF synapses in L2/3 PyCs.** **A** Diagonal transition classes (indicated by the colored arrows) were defined as  $45^\circ$ -wide regions of the  $\Delta NO$  /  $\Delta O$  transitions space, covering  $\pm 22.5^\circ$  around the  $\Delta O = \Delta NO$  and  $\Delta O = -\Delta NO$  diagonal lines. NO (black) and O (red) responses averaged across all cells transitioning along "diagonal" transition classes (indicated by the colored arrows) in Naive and Expert conditions, for the Context-contrasting population and the experimentally-measured population of L2/3 neurons. **B** For simultaneous decrease in NO and increase O responses to familiar images. **C** For simultaneous increase in NO and O responses to novel images.